In [87]:
```
import pandas as pd
import seaborn as sns
```

In [88]:
```
df=pd.read_csv("cleaned file.csv")
```

In [89]:
```
df
```

Out[89]:

| | Unnamed: 0 | country_name | year | life_ladder | log_gdp_per_capita | social_support |
|---|---|---|---|---|---|---|
| **0** | 0 | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 |
| **1** | 1 | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 |
| **2** | 2 | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 |
| **3** | 3 | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 |
| **4** | 4 | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 |
| **...** | ... | ... | ... | ... | ... | ... |
| **1920** | 1944 | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 |
| **1921** | 1945 | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 |
| **1922** | 1946 | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 |
| **1923** | 1947 | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 |
| **1924** | 1948 | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 |

1925 rows × 12 columns

In [90]:
```
df.describe()
```

Out[90]:

| | Unnamed: 0 | year | life_ladder | log_gdp_per_capita | social_support | healt |
|---|---|---|---|---|---|---|
| **count** | 1925.000000 | 1925.000000 | 1925.000000 | 1925.000000 | 1925.000000 | |
| **mean** | 973.437403 | 2013.231169 | 5.468297 | 9.366309 | 0.812817 | |
| **std** | 562.487417 | 4.162258 | 1.118544 | 1.143741 | 0.118451 | |
| **min** | 0.000000 | 2005.000000 | 2.375000 | 6.635000 | 0.290000 | |
| **25%** | 487.000000 | 2010.000000 | 4.640000 | 8.474000 | 0.750000 | |
| **50%** | 974.000000 | 2013.000000 | 5.387000 | 9.444000 | 0.836000 | |
| **75%** | 1462.000000 | 2017.000000 | 6.290000 | 10.335000 | 0.905000 | |
| **max** | 1948.000000 | 2020.000000 | 8.019000 | 11.648000 | 0.987000 | |

In [91]:
```
df.median(numeric_only=True)
```

Out[91]:
```
Unnamed: 0                         974.000
year                              2013.000
life_ladder                          5.387
log_gdp_per_capita                   9.444
social_support                       0.836
healthy_life_expectancy_at_birth    65.000
freedom_to_make_life_choices         0.760
generosity                          -0.016
perceptions_of_corruption            0.794
positive_affect                      0.722
negative_affect                      0.258
dtype: float64
```

In [92]:
```
df.mean(numeric_only=True)
```

Out[92]:
```
Unnamed: 0                          973.437403
year                               2013.231169
life_ladder                           5.468297
log_gdp_per_capita                    9.366309
social_support                        0.812817
healthy_life_expectancy_at_birth     63.353340
freedom_to_make_life_choices          0.742469
generosity                            0.000619
perceptions_of_corruption             0.746940
positive_affect                       0.710019
negative_affect                       0.268366
dtype: float64
```

In [93]:
```
df["life_ladder"].quantile([0.25, 0.5, 0.75])
```

Out[93]:
```
0.25    4.640
0.50    5.387
0.75    6.290
Name: life_ladder, dtype: float64
```

In [94]:
```
df.quantile([0.25, 0.5, 0.75],numeric_only=True)
```

Out[94]:

| | Unnamed: 0 | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_exp |
|---|---|---|---|---|---|---|
| **0.25** | 487.0 | 2010.0 | 4.640 | 8.474 | 0.750 | |
| **0.50** | 974.0 | 2013.0 | 5.387 | 9.444 | 0.836 | |
| **0.75** | 1462.0 | 2017.0 | 6.290 | 10.335 | 0.905 | |

In [95]:
```
df = df.drop("Unnamed: 0", axis=1)
```

NORMAL DISTRIBUTION

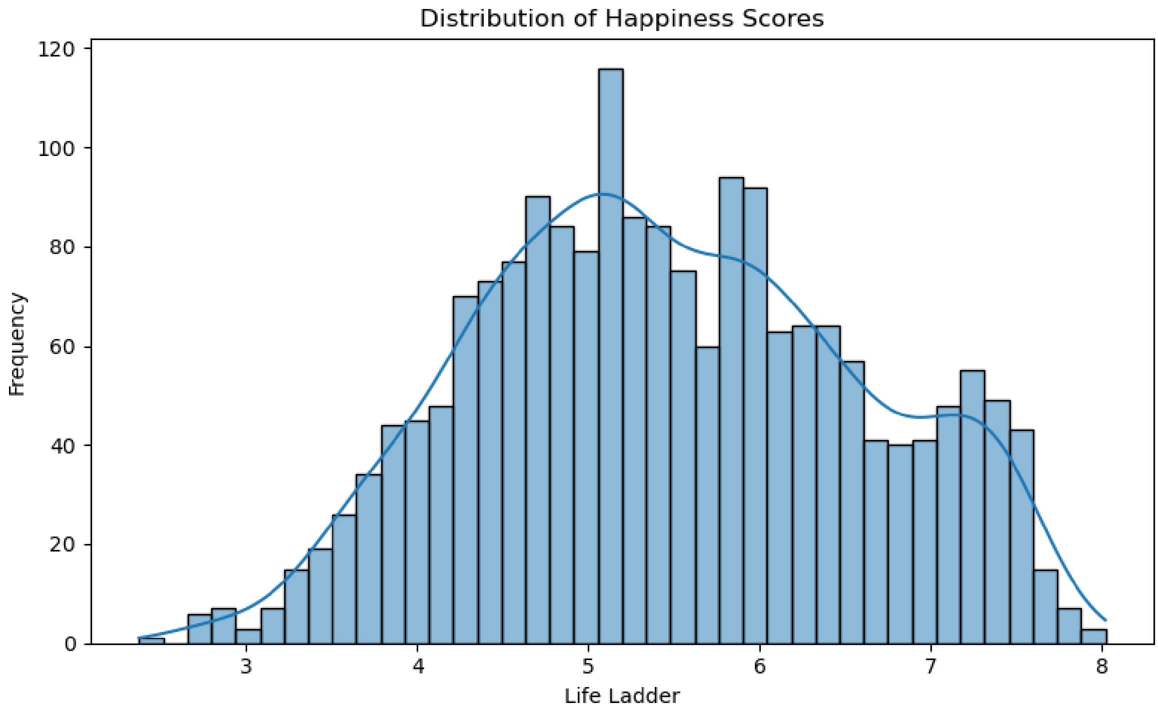
In [96]:
```
df.value_counts()
```

Out[96]:

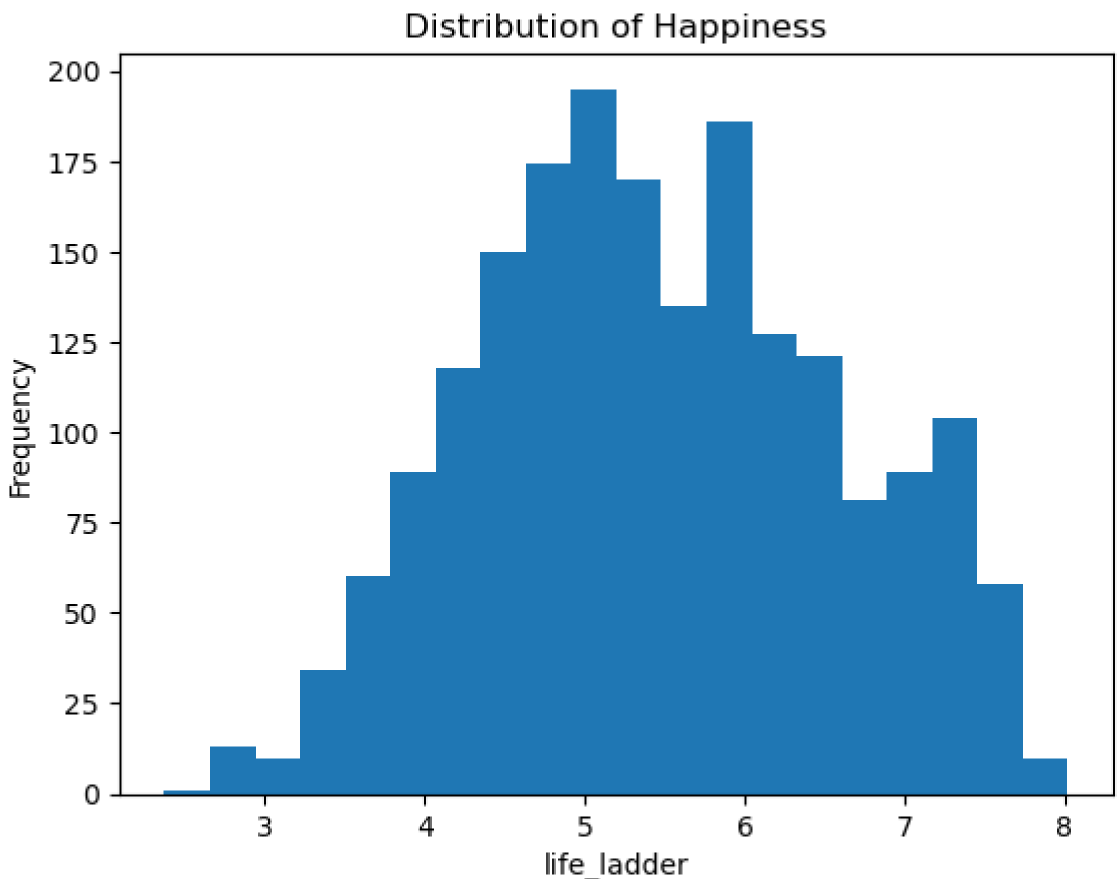
```
country_name  year  life_ladder  log_gdp_per_capita  social_support  healthy_li
fe_expectancy_at_birth  freedom_to_make_life_choices  generosity  perceptions_o
f_corruption  positive_affect  negative_affect
Afghanistan   2008  3.724        7.370              0.451           50.80
0.718                         0.168      0.882                      0.518
0.258              1
Netherlands   2014  7.321        10.867             0.909           71.88
0.910                         0.331      0.457                      0.868
0.221              1
Norway        2017  7.579        11.050             0.950           73.10
0.953                         0.236      0.250                      0.849
0.203              1
              2016  7.596        11.035             0.960           73.00
0.954                         0.133      0.410                      0.850
0.209              1
              2015  7.603        11.033             0.947           72.90
0.948                         0.257      0.299                      0.843
0.209              1
                                                                                     
..
Greece        2014  4.756        10.257             0.832           71.74
0.369                        -0.288      0.930                      0.695
0.385              1
              2013  4.720        10.243             0.687           71.68
0.426                        -0.272      0.941                      0.689
0.482              1
              2012  5.096        10.268             0.812           71.62
0.373                        -0.305      0.959                      0.581
0.352              1
              2011  5.372        10.339             0.852           71.56
0.528                        -0.316      0.941                      0.591
0.323              1
Zimbabwe      2020  3.160        7.829              0.717           56.80
0.643                        -0.009      0.789                      0.703
0.346              1
Name: count, Length: 1925, dtype: int64
```

In [97]:

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8,5))
sns.histplot(df["life_ladder"], bins=40, kde=True)
plt.title("Distribution of Happiness Scores")
plt.xlabel("Life Ladder")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```

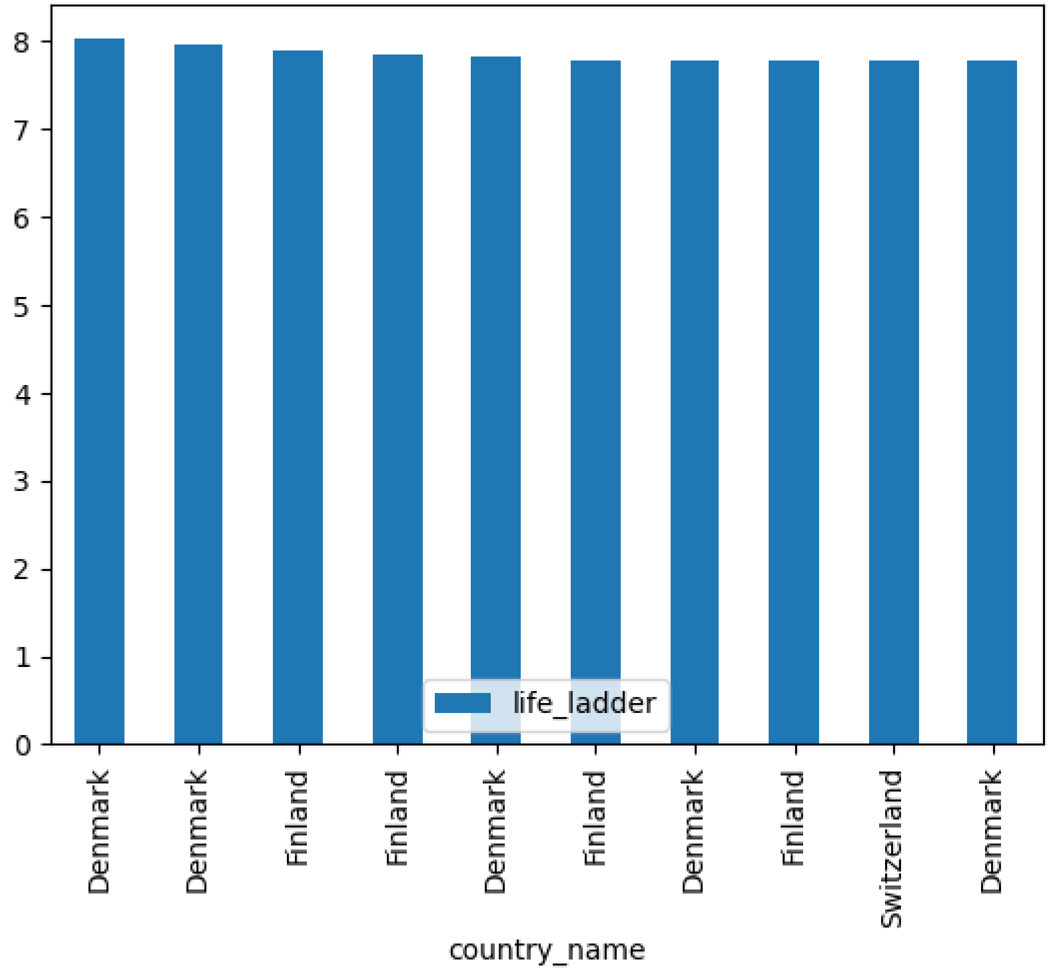



In [98]:
```
import matplotlib.pyplot as plt

plt.hist(df["life_ladder"], bins=20)
plt.xlabel("life_ladder")
plt.ylabel("Frequency")
plt.title("Distribution of Happiness")
plt.show()
```



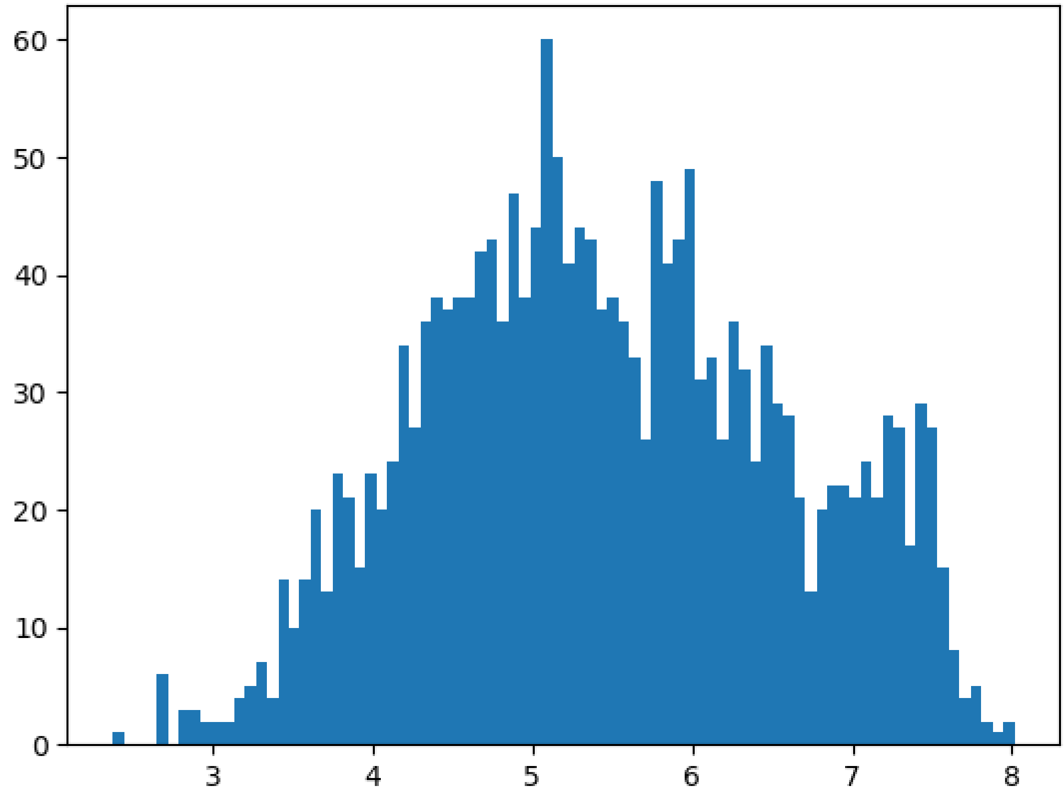

In [99]:
```
df.sort_values("life_ladder", ascending=False).head(10).plot(x="country_name", y
```

Out[99]: <Axes: xlabel='country_name'>




In [100…

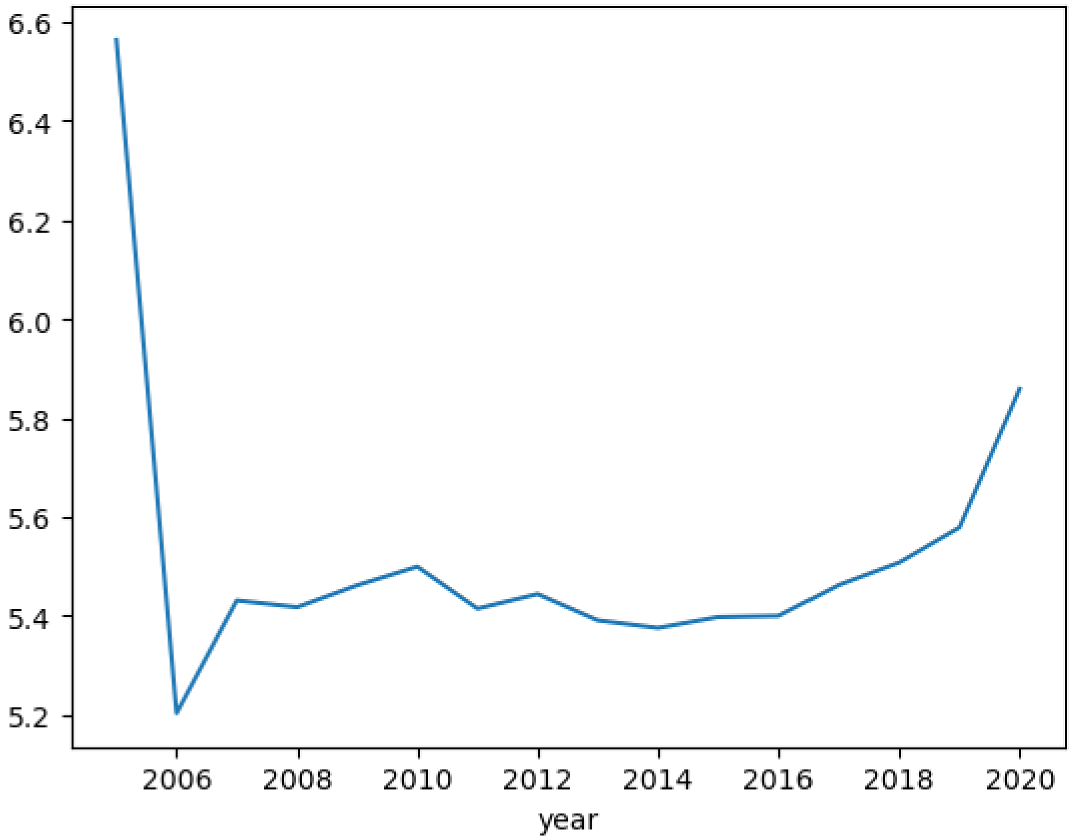
```
plt.hist(df["life_ladder"], bins=82)
plt.show()
```



HAPPINESS OVER YEARS

In [101…
```
df.groupby("year")["life_" \
"ladder"].mean().plot()
```

Out[101…
```
<Axes: xlabel='year'>
```




Happiness levels decreased sharply in the early years, remained relatively stable for a decade, and showed gradual improvement in recent years.

In [102…
```
df["life_ladder"].skew()
```

Out[102…
```
np.float64(0.06466158095247462)
```

In [103…
```
df["healthy_life_expectancy_at_birth"].skew()
```

Out[103…
```
np.float64(-0.7514343287731668)
```

In [104…
```
df
```

Out[104…

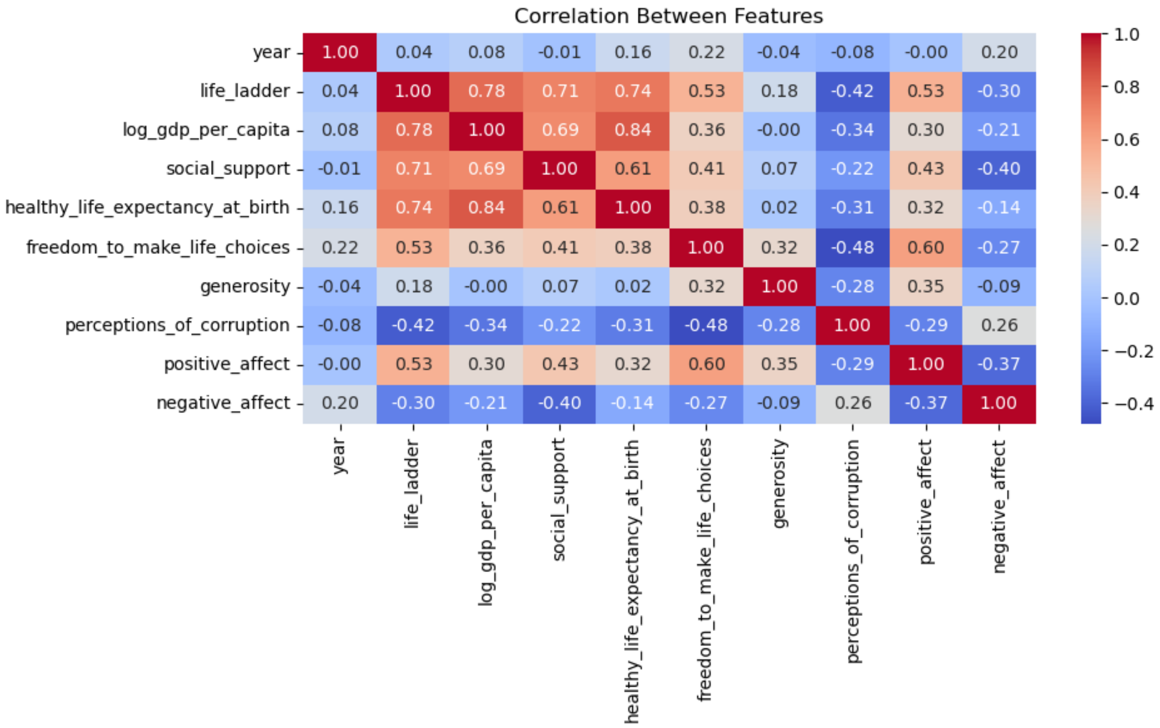
| | country_name | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_ |
|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | |
| **...** | ... | ... | ... | ... | ... | |
| **1920** | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 | |
| **1921** | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 | |
| **1922** | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 | |
| **1923** | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 | |
| **1924** | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 | |

1925 rows × 11 columns

### Correlation

In [105…

```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, fmt=".2f", cmap="coolwarm")
plt.title("Correlation Between Features")
plt.tight_layout()
plt.show()
```




### Insight

- Strong Positive Relationships (+):life_ladder ↔ log_gdp_per_capita : ~0.78, life_ladder

↔ social_support : ~0.71, life_ladder ↔ healthy_life_expectancy_at_birth : ~0.74, log_gdp_per_capita ↔ healthy_life_expectancy_at_birth : ~0.84, social_support ↔ log_gdp_per_capita : ~0.69, freedom_to_make_life_choices ↔ positive_affect : ~0.60

🔹 Strong Negative Relationships (−):life_ladder ↔ perceptions_of_corruption : ~ -0.42, freedom_to_make_life_choices ↔ perceptions_of_corruption : ~ -0.48, social_support ↔ negative_affect : ~ -0.40, positive_affect ↔ negative_affect : ~ -0.37

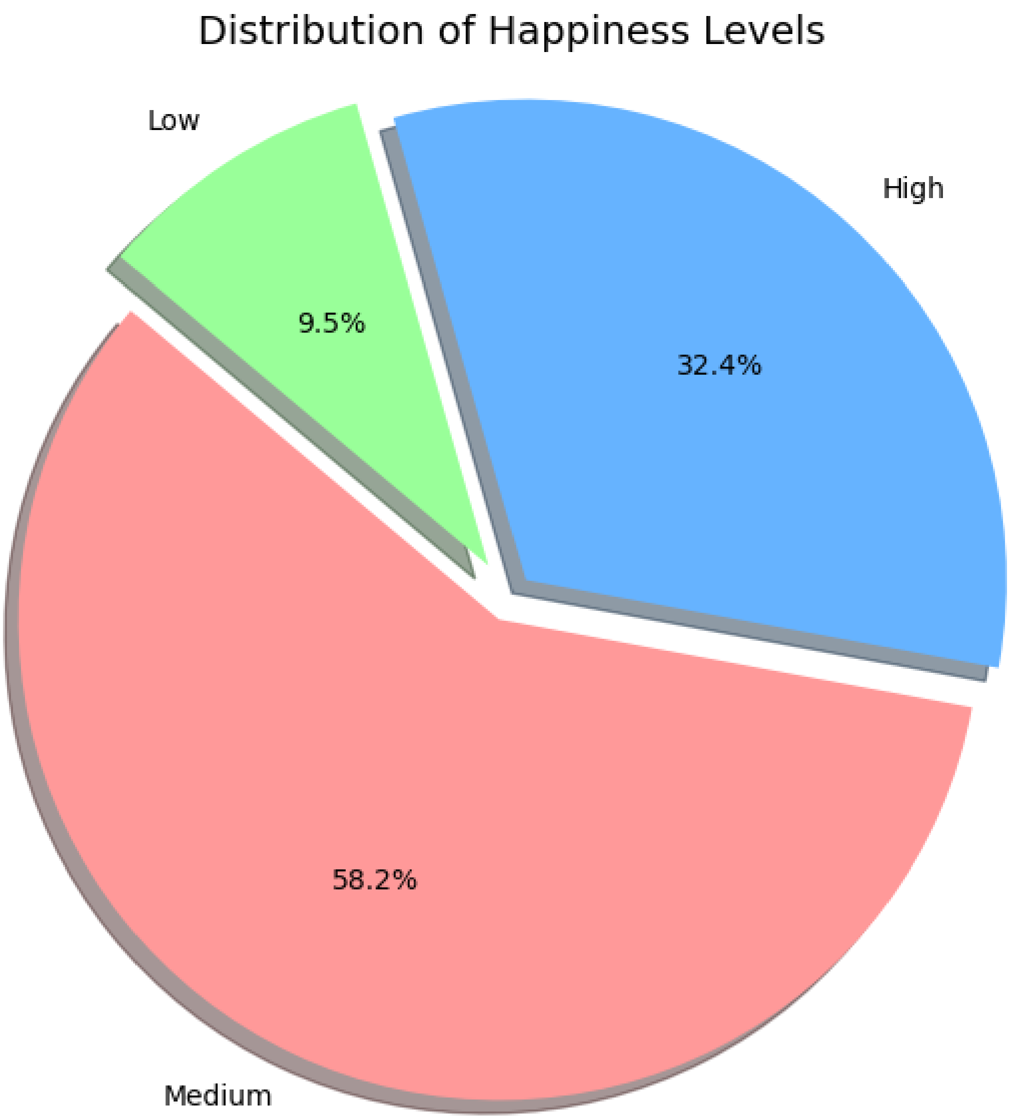
🔹 Moderate Relationships:life_ladder ↔ freedom_to_make_life_choices : ~0.53, life_ladder ↔ positive_affect : ~0.53, log_gdp_per_capita ↔ social_support : ~0.69, healthy_life_expectancy_at_birth ↔ social_support : ~0.61, generosity ↔ positive_affect : ~0.35

🔹 Weak / No Relationships:year ↔ most variables : very weak correlation, generosity ↔ most variables : weak correlation, log_gdp_per_capita ↔ generosity : ~0.00 (no relation) Globally, happiness (life_ladder) is strongly positively related to GDP, social support, and life expectancy, while it is negatively affected by corruption and negative emotions.

**Pie Chart**

In [106…

```
import matplotlib.pyplot as plt
import pandas as pd

df["happiness_level"] = pd.cut(
    df["life_ladder"],
    bins=[0, 4, 6, 10],
    labels=["Low", "Medium", "High"]
)

counts = df["happiness_level"].value_counts()

plt.figure(figsize=(7,7))

colors = ["#ff9999", "#66b3ff", "#99ff99"]
explode = (0.05, 0.05, 0.08)

plt.pie(
    counts,
    labels=counts.index,
    autopct="%1.1f%%",
    startangle=140,
    colors=colors,
    explode=explode,
    shadow=True
)

plt.title("Distribution of Happiness Levels", fontsize=14)
plt.axis("equal")

plt.show()
```




Majority of countries fall in the Medium happiness category. A significant portion has High happiness levels. Very few countries are in the Low happiness category. Happiness distribution is skewed towards moderate levels

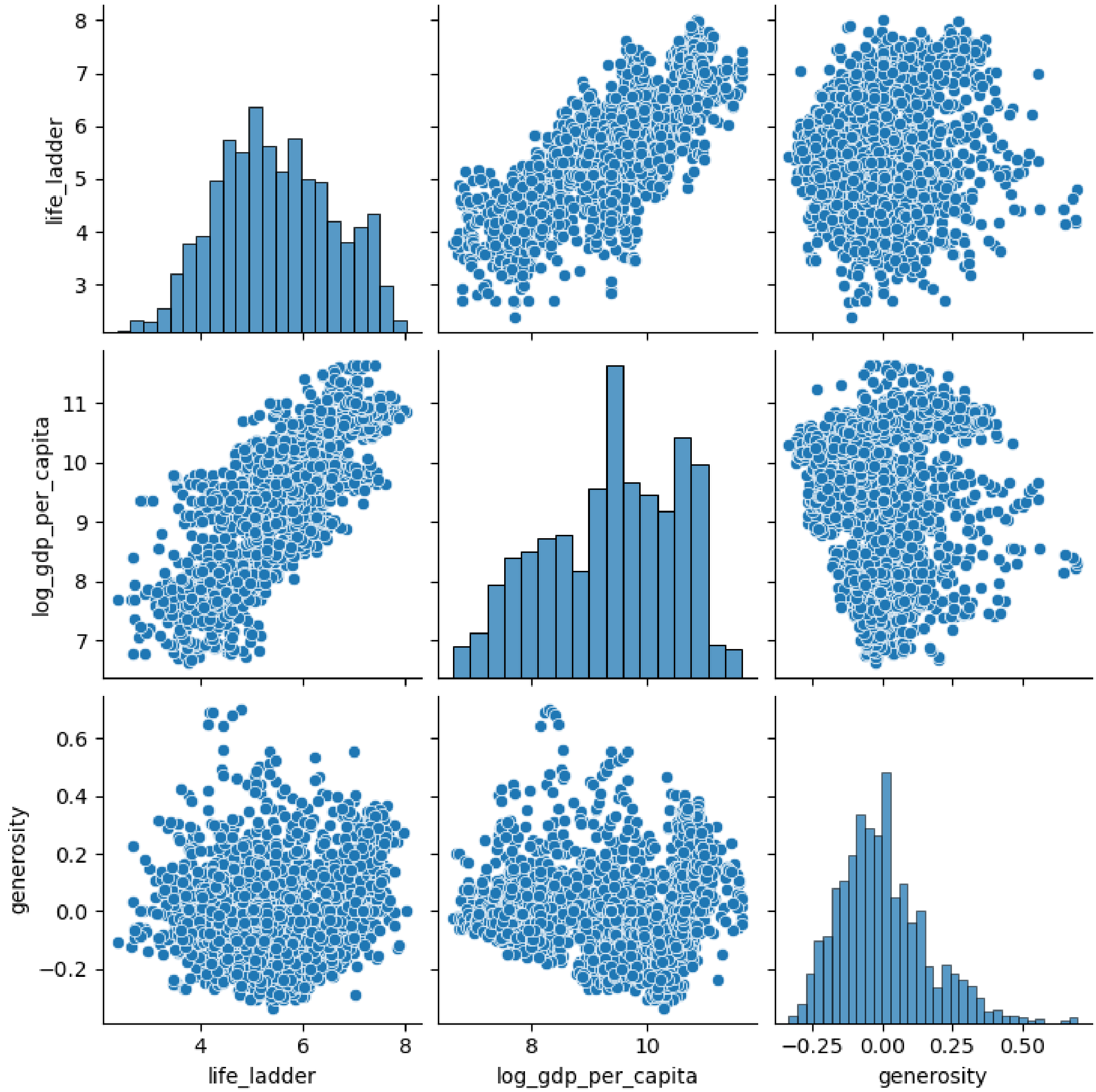
RELATION

In [107…

```
import seaborn as sns
sns.pairplot(df[["life_ladder", "log_gdp_per_capita", "generosity"]])
```

Out[107…

```
<seaborn.axisgrid.PairGrid at 0x262e2031a90>
```



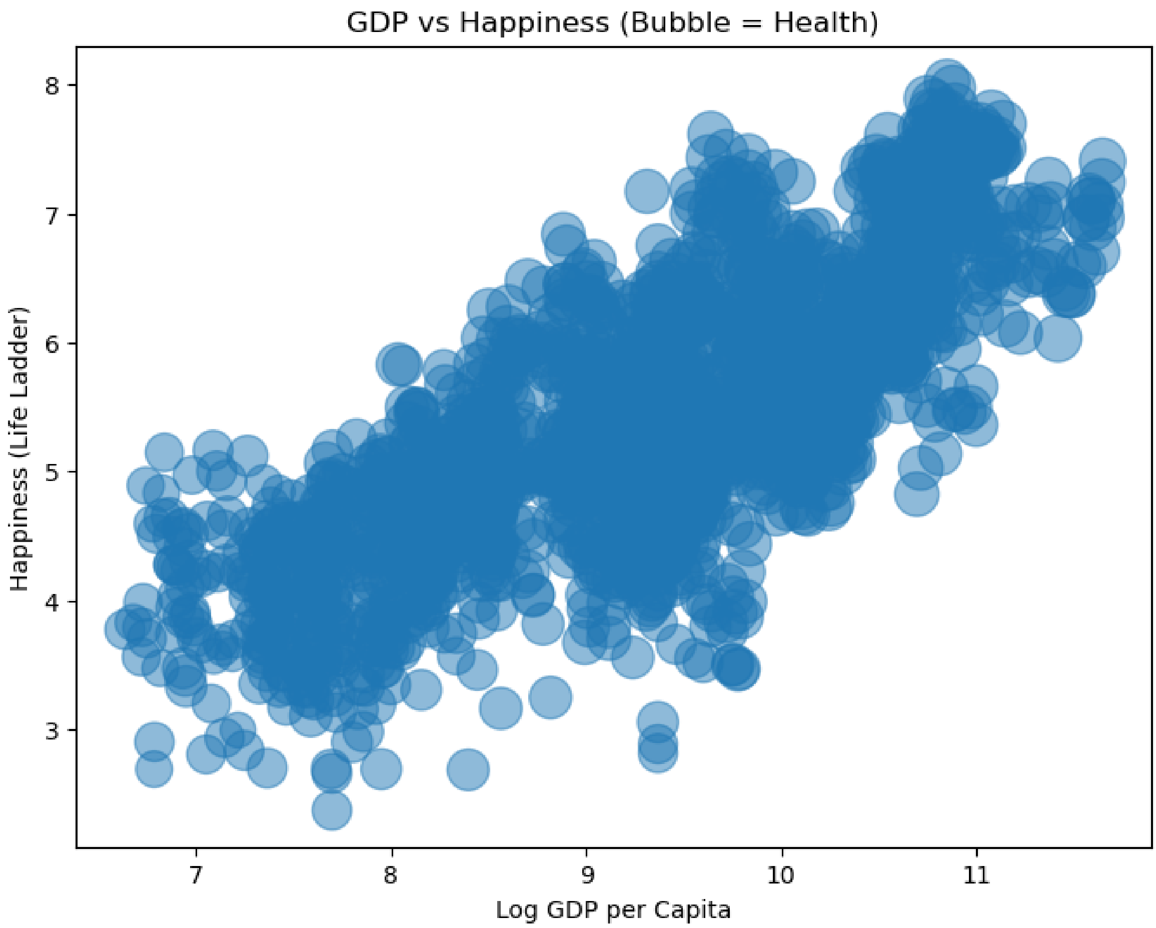

In [108…

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

sns.regplot(
    data=df,
    x="healthy_life_expectancy_at_birth",
    y="life_ladder"
)

plt.title("Happiness vs Health")
plt.xlabel("Healthy Life Expectancy")
plt.ylabel("Life Ladder (Happiness)")

plt.tight_layout()
plt.show()
```


happiness and health is directly proportional

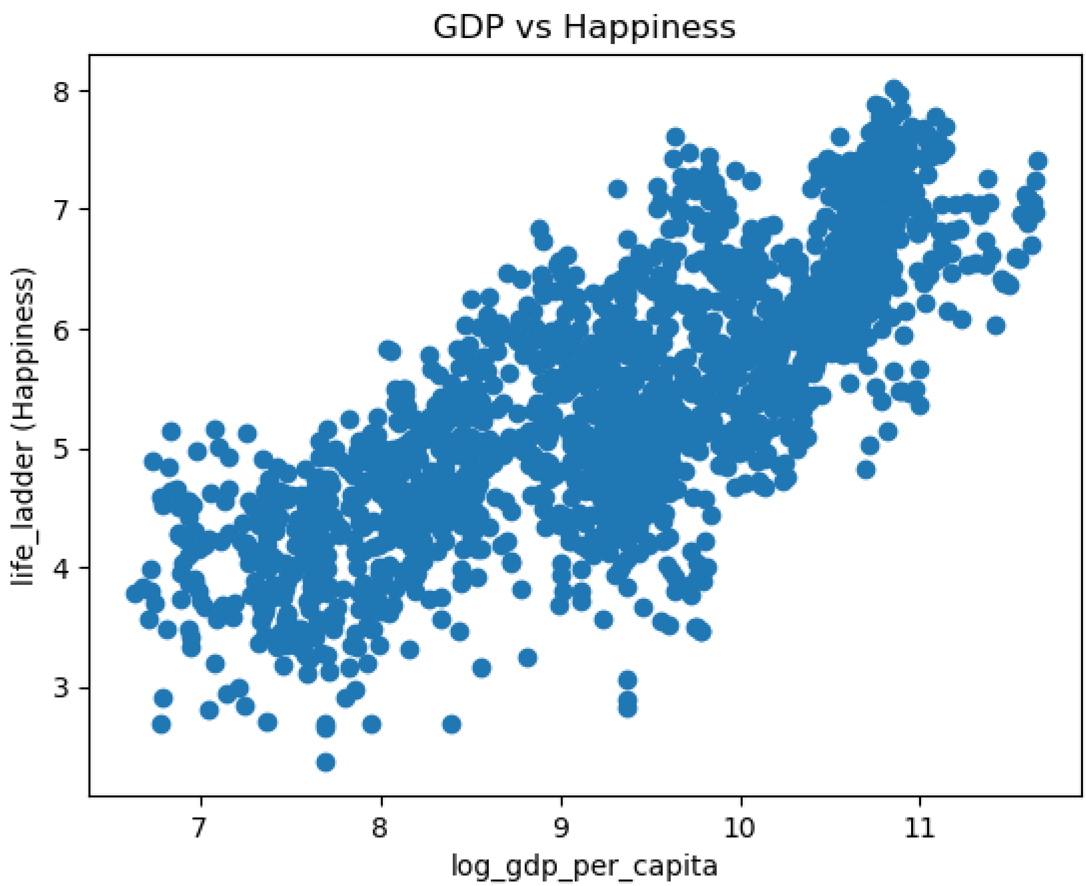
In [109…

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))

plt.scatter(
    df["log_gdp_per_capita"],
    df["life_ladder"],
    s=df["healthy_life_expectancy_at_birth"] * 5,
    alpha=0.5
)

plt.xlabel("Log GDP per Capita")
plt.ylabel("Happiness (Life Ladder)")
plt.title("GDP vs Happiness (Bubble = Health)")

plt.show()
```




In [110…

```
import matplotlib.pyplot as plt
plt.scatter(df["log_gdp_per_capita"], df["life_ladder"])
plt.xlabel("log_gdp_per_capita")
plt.ylabel("life_ladder (Happiness)")
plt.title("GDP vs Happiness")
plt.show()
```

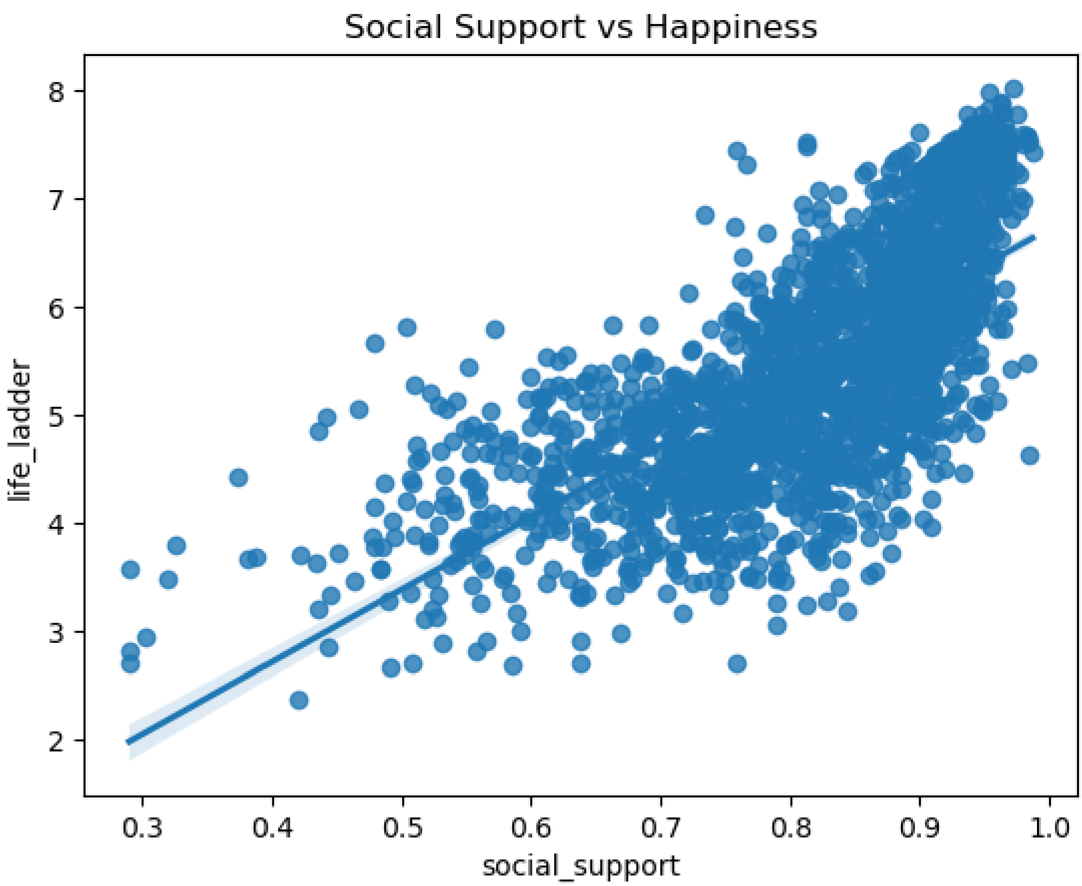



In [111…
```
sns.kdeplot(
    data=df,
    x="healthy_life_expectancy_at_birth",
    y="life_ladder",
    fill=True
)

plt.title("Density: Happiness vs Health")
plt.show()
```


gdp and happiness are directly proportional

In [112…

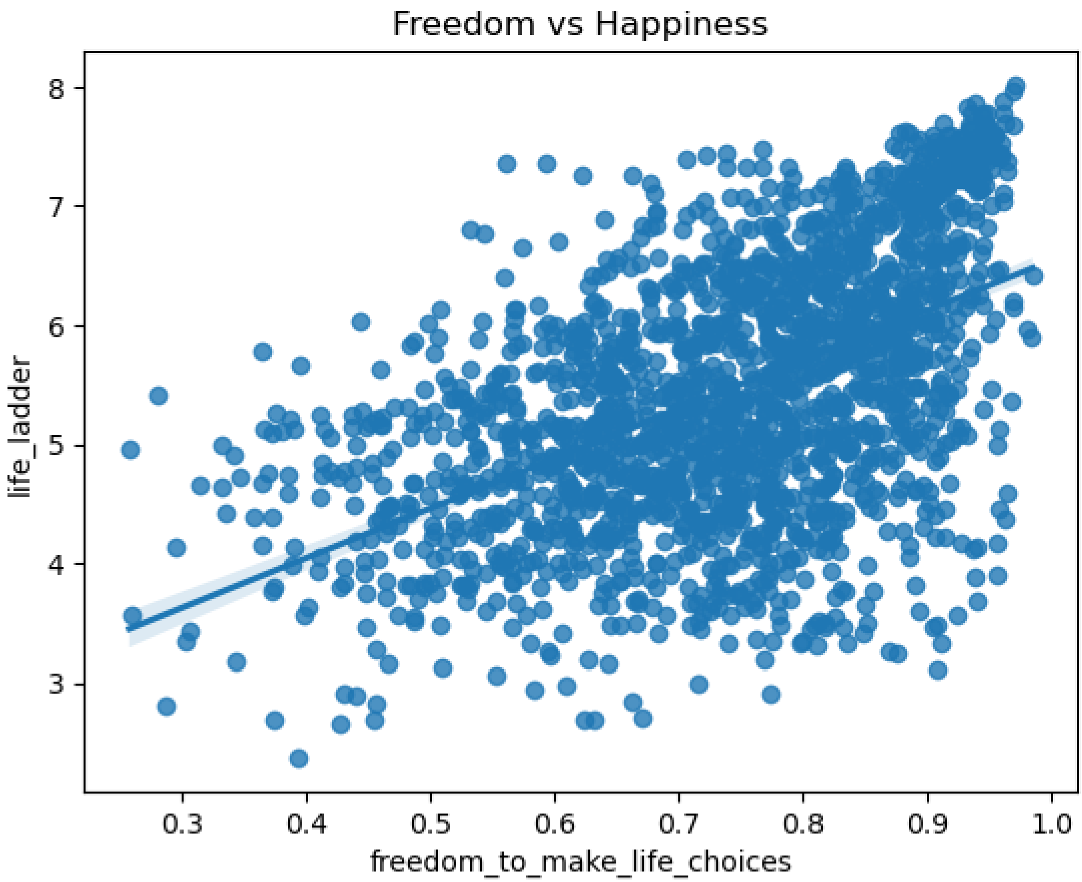
```
sns.regplot(data=df, x="social_support", y="life_ladder")
plt.title("Social Support vs Happiness")
plt.show()
```




Social support and happiness directly proportional

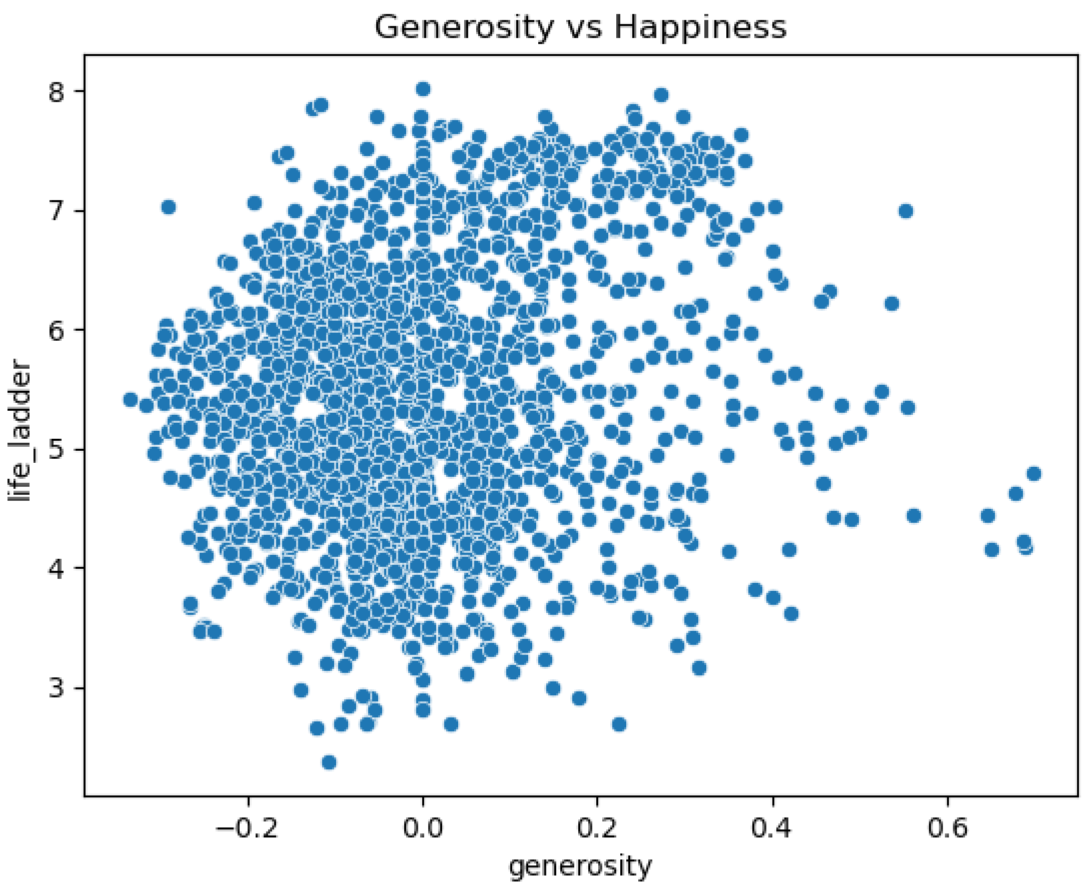
In [113…

```
sns.regplot(data=df, x="freedom_to_make_life_choices", y="life_ladder")
plt.title("Freedom vs Happiness")
plt.show()
```




freedom and happiness is directly proportional

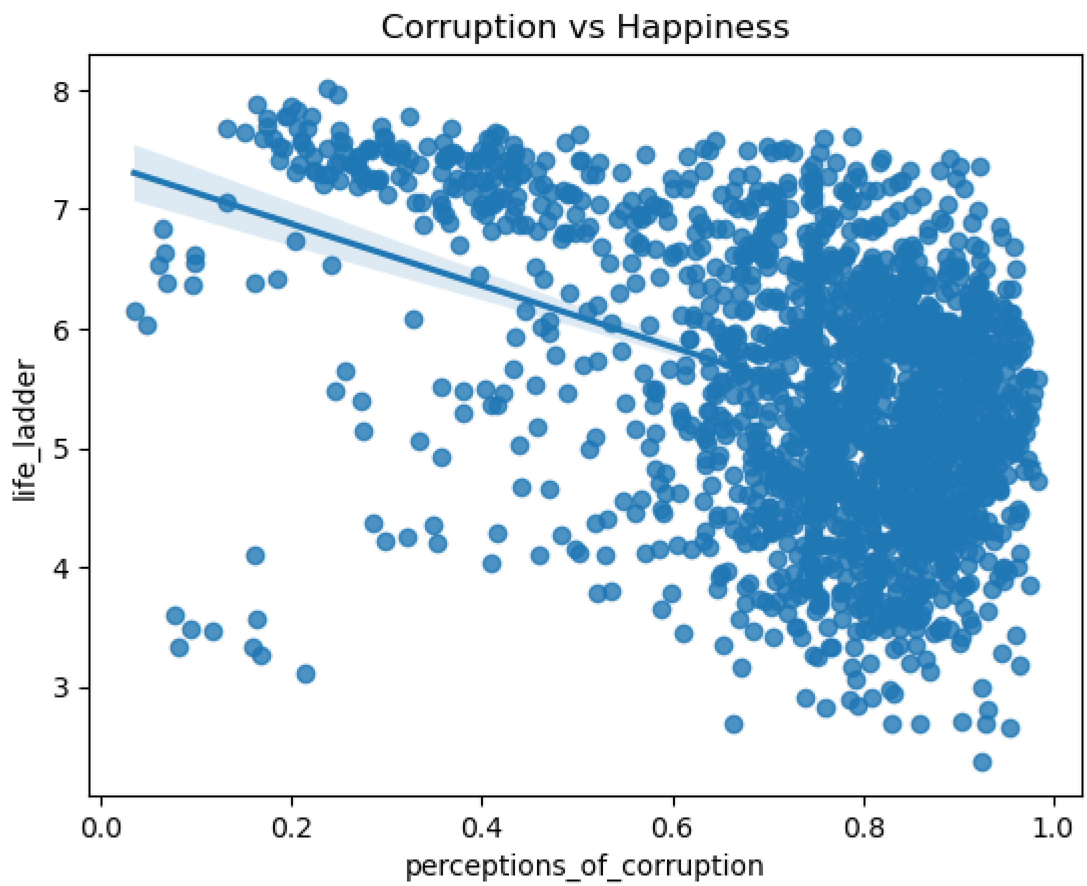
In [114…
```
sns.scatterplot(data=df, x="generosity", y="life_ladder")
plt.title("Generosity vs Happiness")
plt.show()
```




there is no proportionality between genorosity and happiness

In [115…

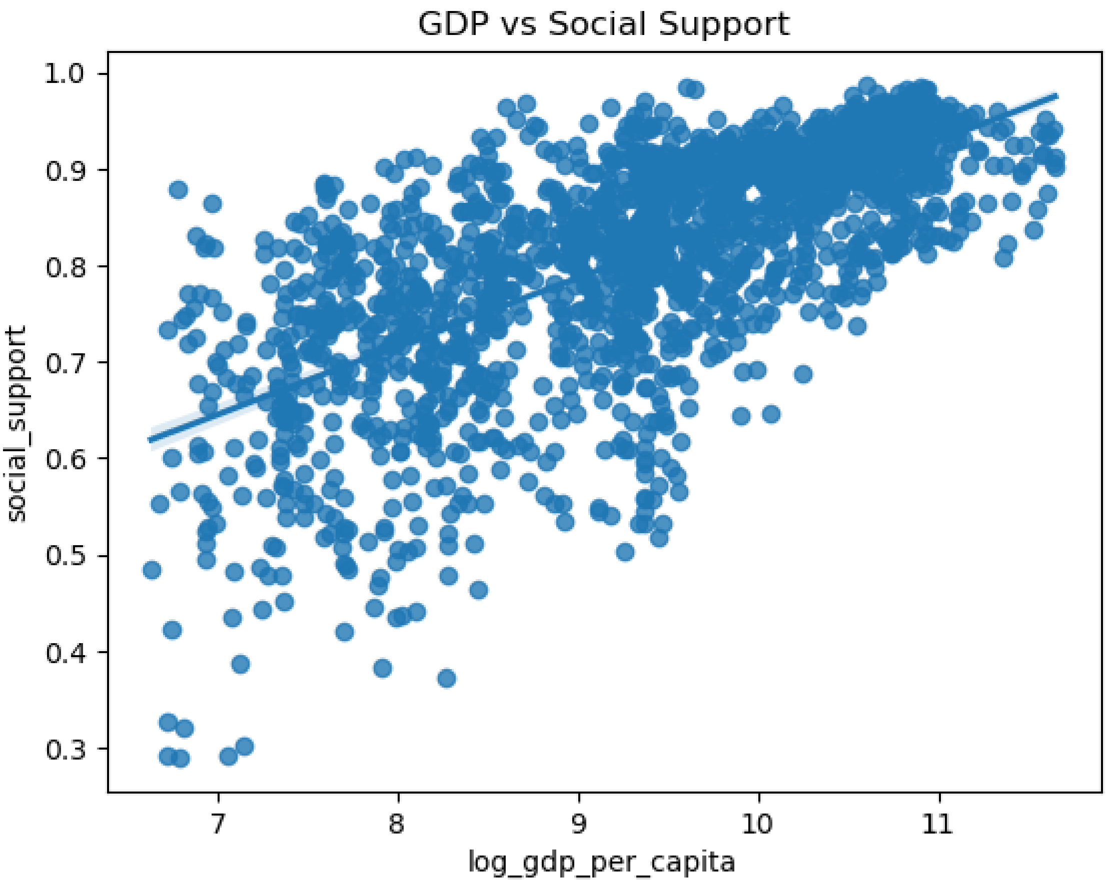
```
sns.regplot(data=df, x="perceptions_of_corruption", y="life_ladder")
plt.title("Corruption vs Happiness")
plt.show()
```




here happiness is inversely propotional to corruption

In [116…

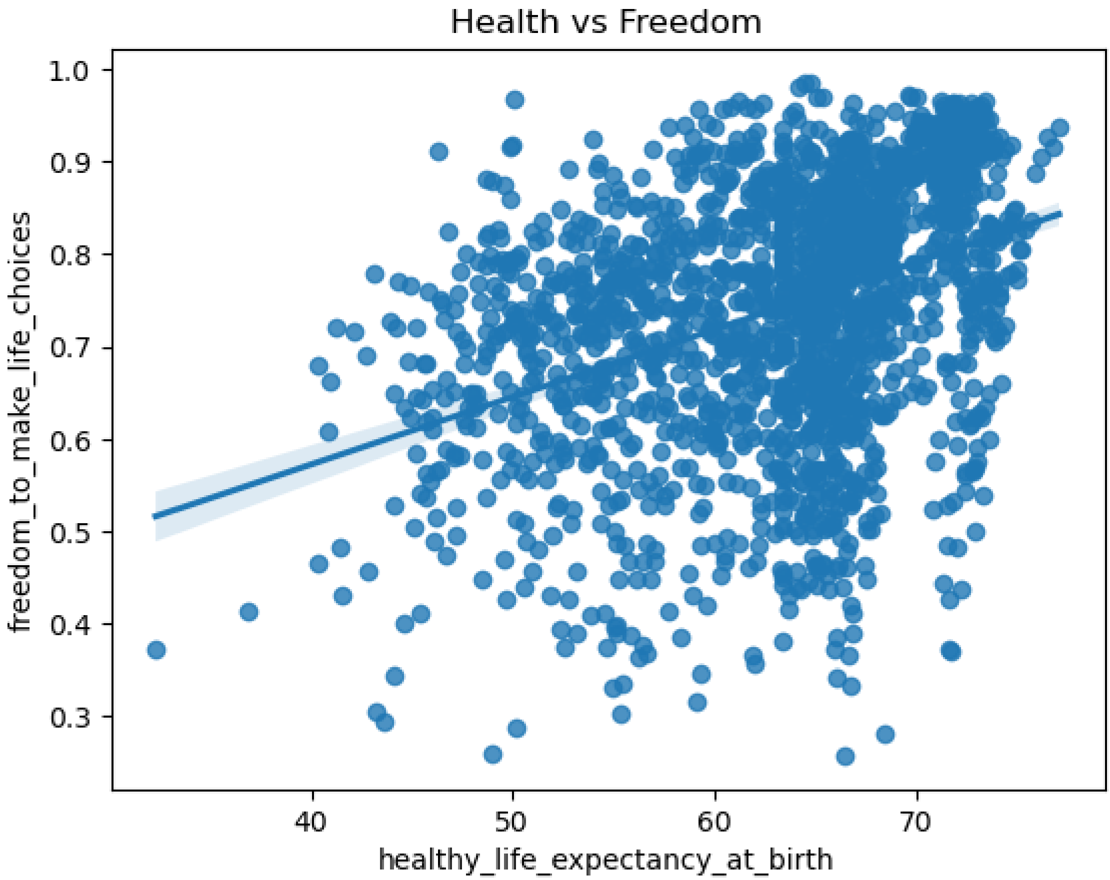
```
sns.regplot(data=df, x="log_gdp_per_capita", y="social_support")
plt.title("GDP vs Social Support")
plt.show()
```




gdp and social support is directly propotional

In [117…

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.regplot(
    data=df,
    x="healthy_life_expectancy_at_birth",
    y="freedom_to_make_life_choices"
)

plt.title("Health vs Freedom")
plt.show()
```



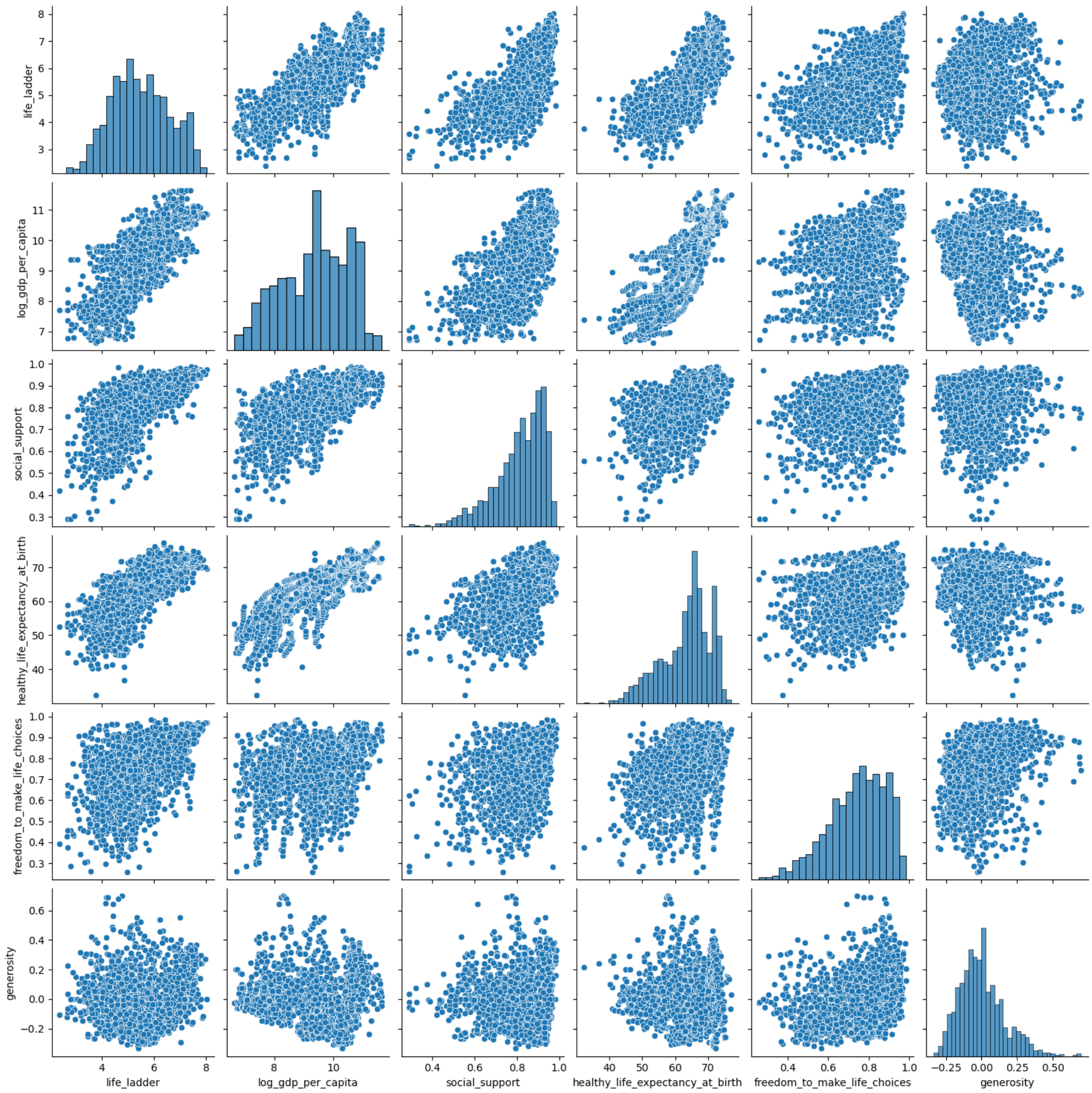

health and freedom is directly propotional

In [118…

```
sns.pairplot(df[[
    "life_ladder",
    "log_gdp_per_capita",
    "social_support",
    "healthy_life_expectancy_at_birth",
    "freedom_to_make_life_choices",
    "generosity"
]])
```

Out[118…

```
<seaborn.axisgrid.PairGrid at 0x262db9008a0>
```

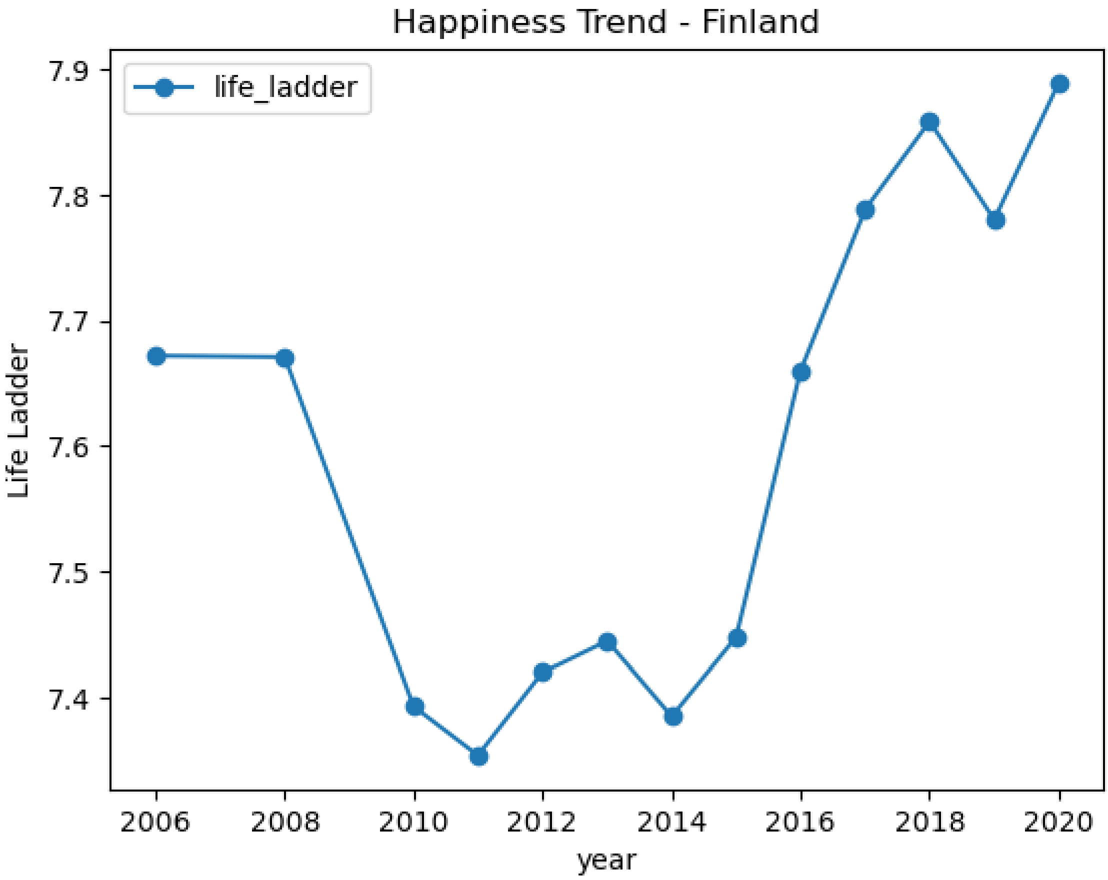



All in one

### 🌍 Country-wise Analysis

In [119…

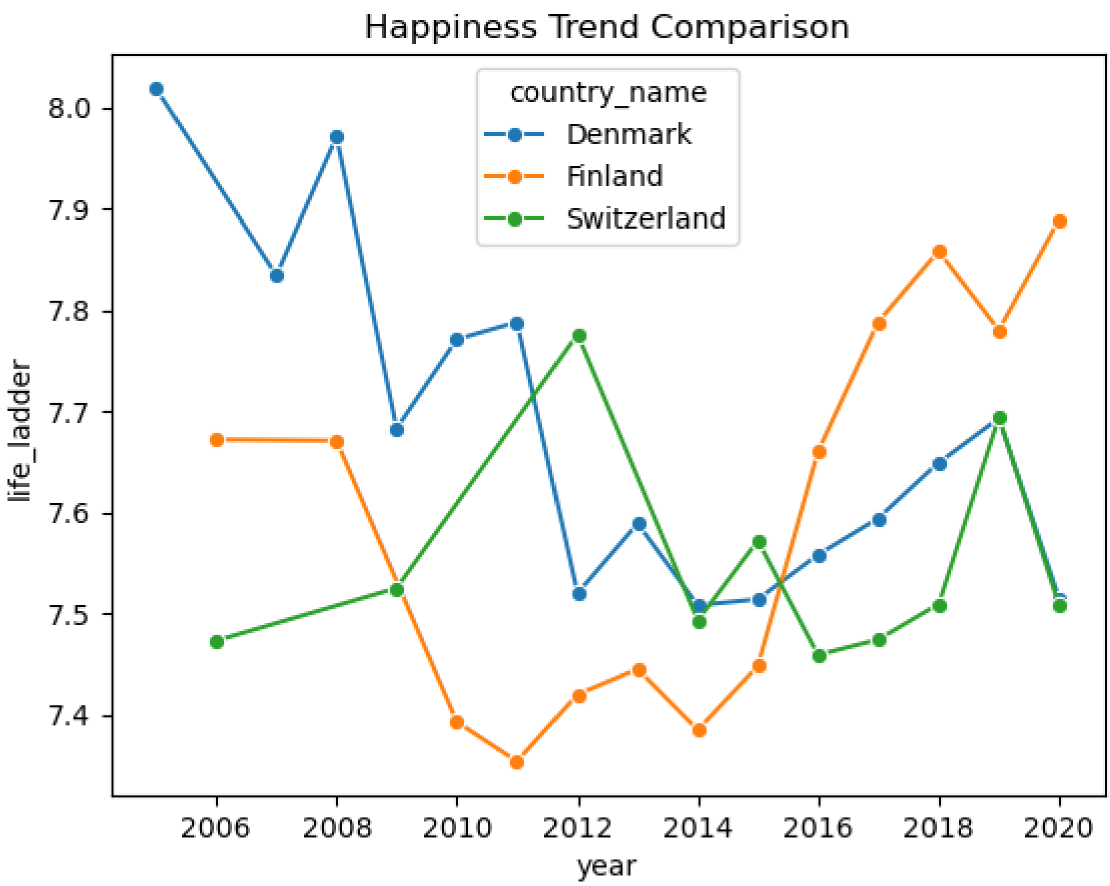
```
country_data = df[df["country_name"] == "Finland"]

country_data.sort_values("year").plot(
    x="year", y="life_ladder", kind="line", marker="o"
)

import matplotlib.pyplot as plt
plt.title("Happiness Trend - Finland")
plt.ylabel("Life Ladder")
plt.show()
```




The happiness trend in finland over the years is increasing

In [120…

```
countries = ["Finland", "Denmark", "Switzerland"]

subset = df[df["country_name"].isin(countries)]

import seaborn as sns

sns.lineplot(
    data=subset,
    x="year",
    y="life_ladder",
    hue="country_name",
    marker="o"
)

plt.title("Happiness Trend Comparison")
plt.show()
```




the happiness trend in some countries over years. the happiness trend Denmark decreasing over years. the happiness trend in Finland increasing over years. the happiness trend in decreasing and increasing over years.

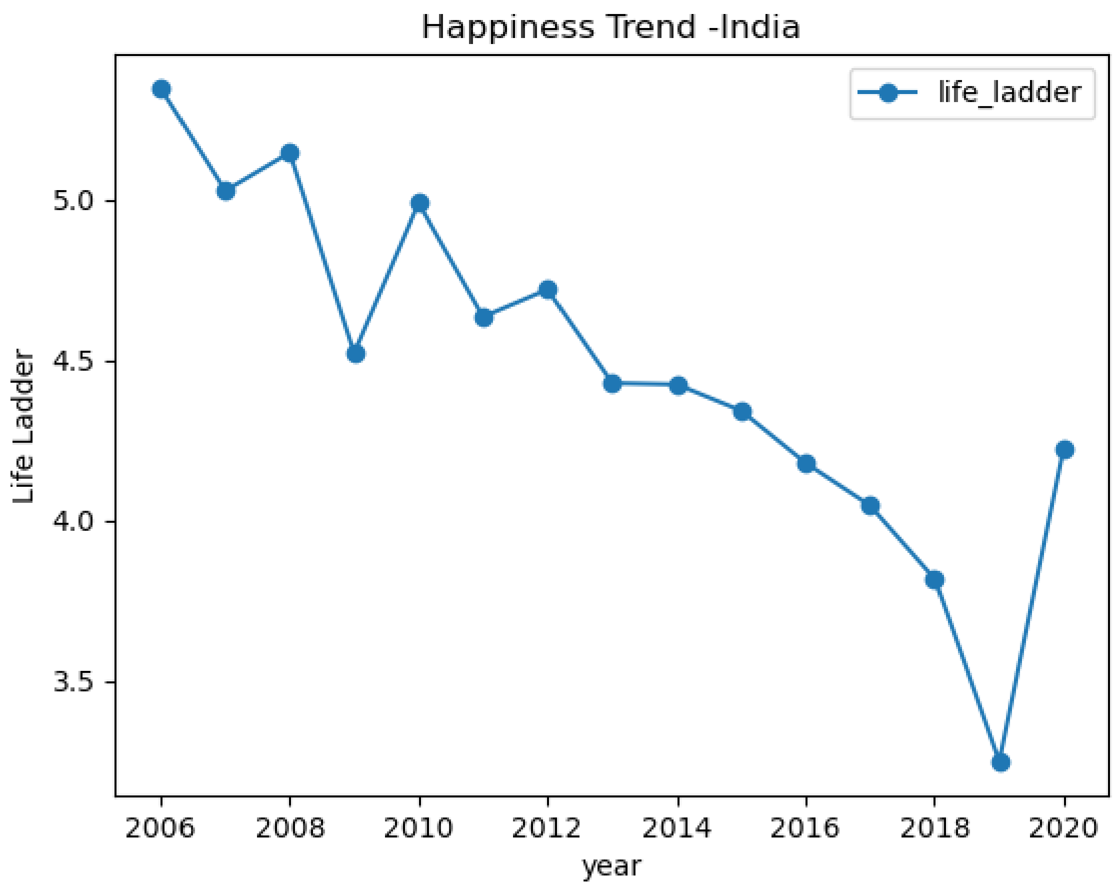
In [121…
```
afg_df = df[df["country_name"] == "Afghanistan"]
```

In [122…
```
afg_df.head()
```

Out[122…

| | country_name | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_exp |
|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | |

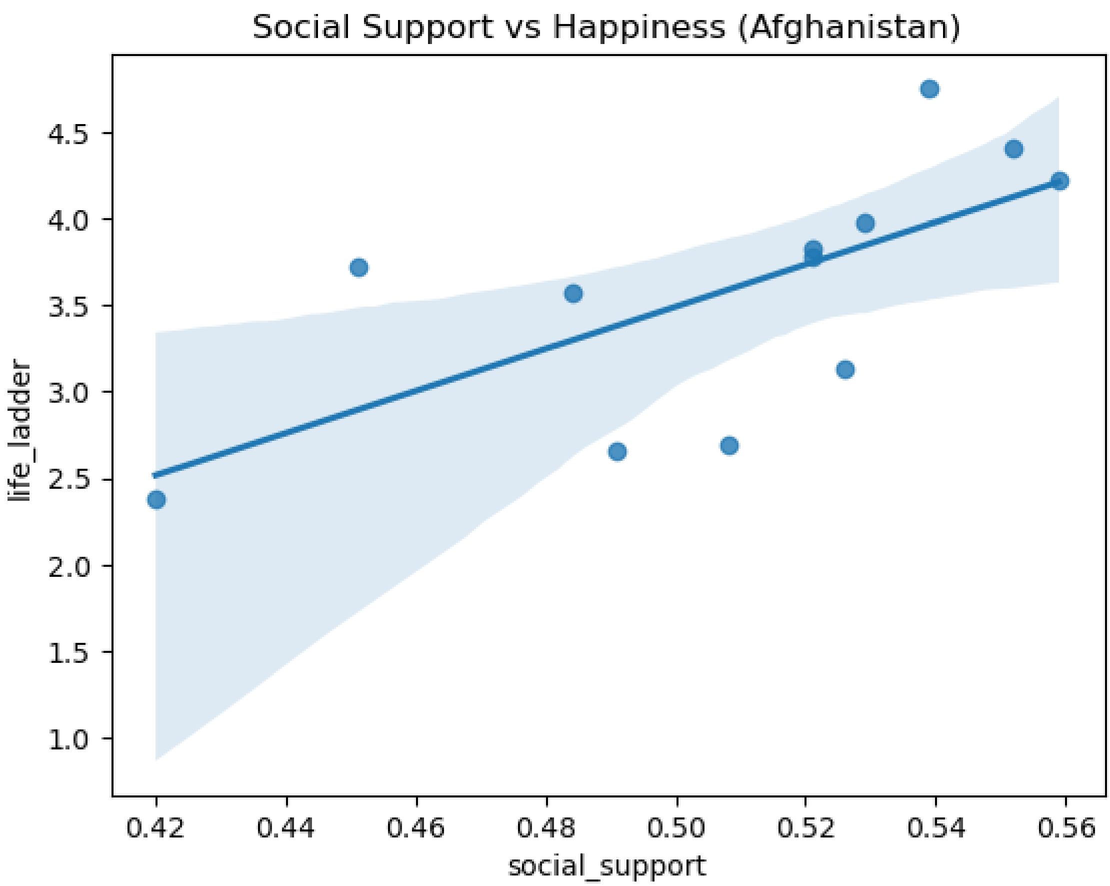
In [123…
```
country_data = df[df["country_name"] == "India"]

country_data.sort_values("year").plot(
    x="year", y="life_ladder", kind="line", marker="o"
)

import matplotlib.pyplot as plt
plt.title("Happiness Trend -India")
plt.ylabel("Life Ladder")
plt.show()
```




happiness trent in india over the years

In [124…

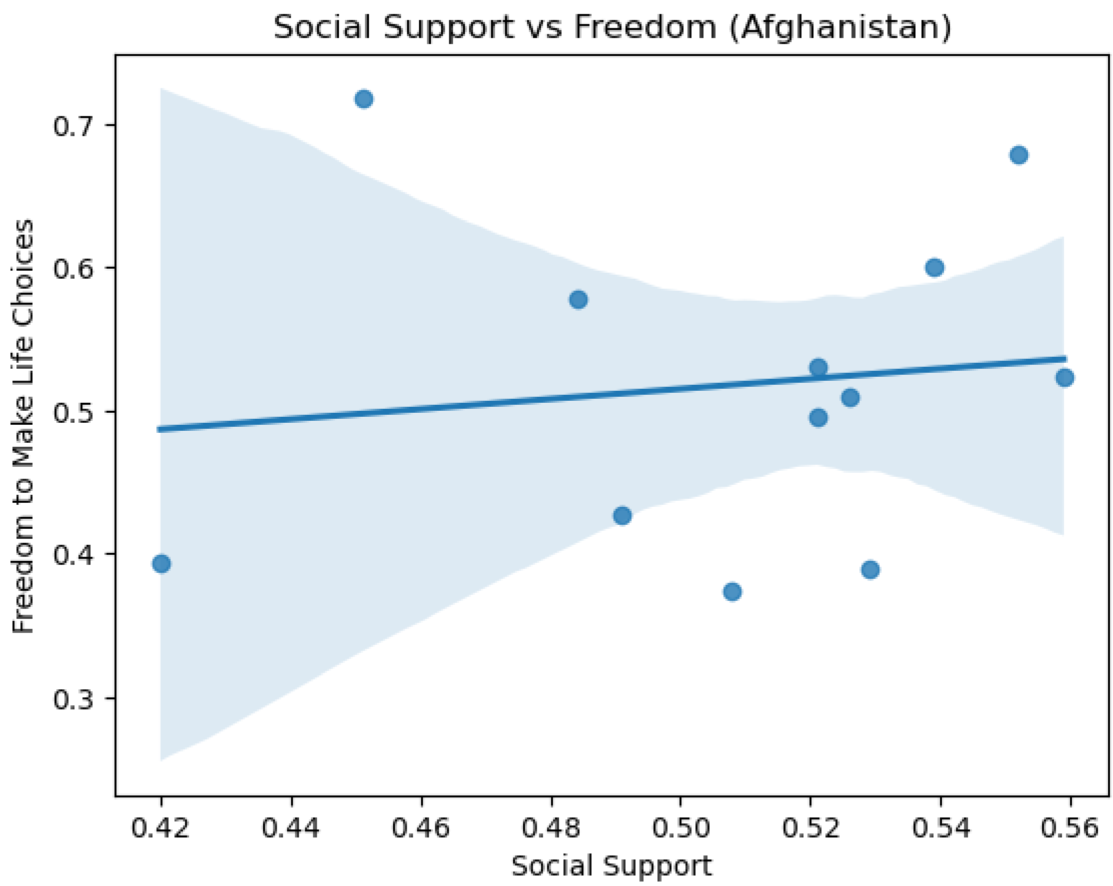
```
sns.regplot(
data=afg_df,
x="social_support",
y="life_ladder"
)
plt.title("Social Support vs Happiness (Afghanistan)")
plt.show()
```




the relation btw happiness and social support in Afghanisthan is directly propotional

In [125…

```
sns.regplot(
data=afg_df,
x="social_support",
y="freedom_to_make_life_choices"
)
plt.title("Social Support vs Freedom (Afghanistan)")
plt.xlabel("Social Support")
plt.ylabel("Freedom to Make Life Choices")
plt.show()
```

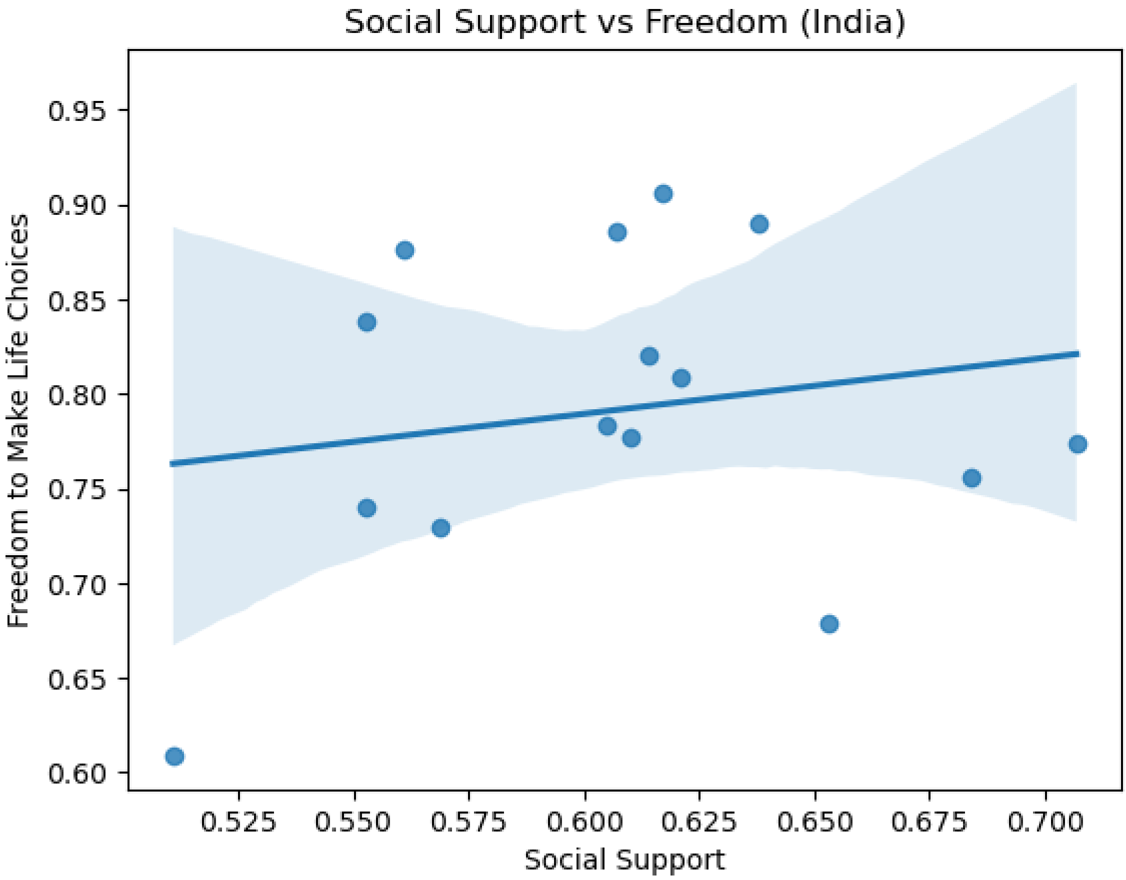



In [126…

```
ind_df = df[df["country_name"] == "India"]
```

In [127…

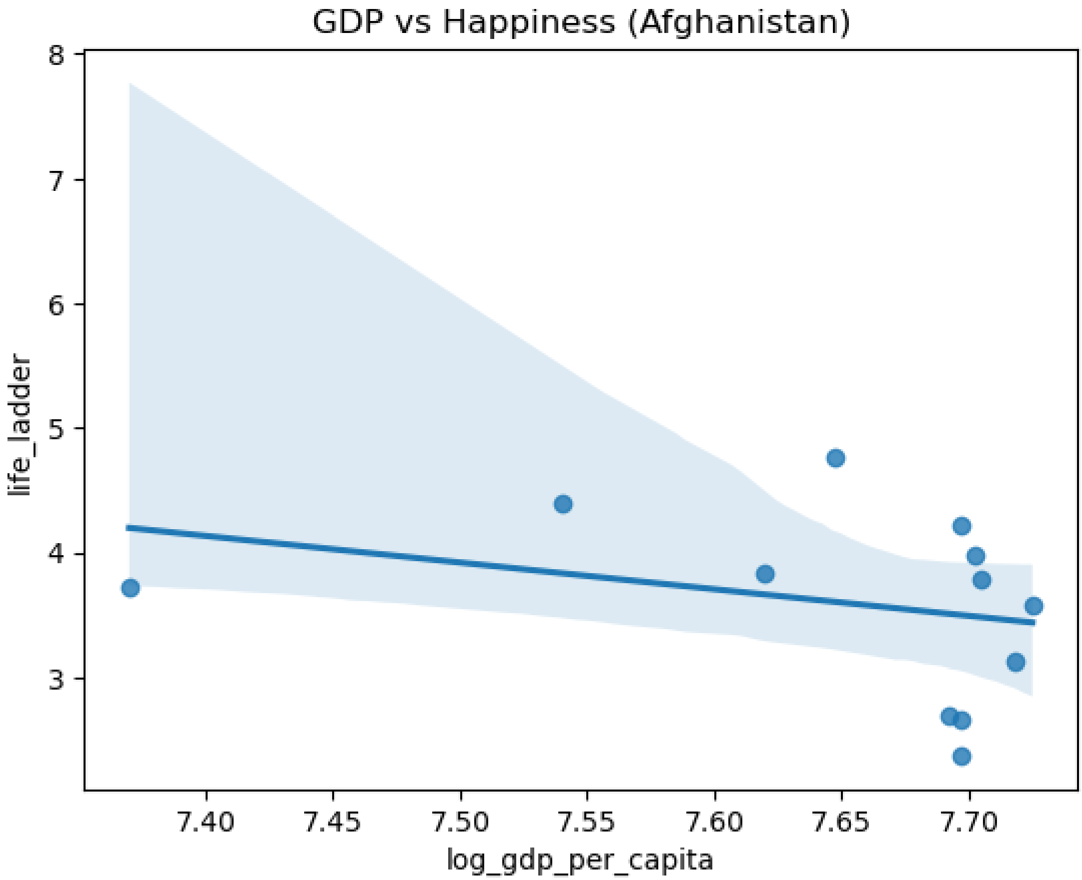
```
sns.regplot(
data=ind_df,
x="social_support",
y="freedom_to_make_life_choices"
)
plt.title("Social Support vs Freedom (India)")
plt.xlabel("Social Support")
plt.ylabel("Freedom to Make Life Choices")
plt.show()
```




the relation btw Freedom and social support in India

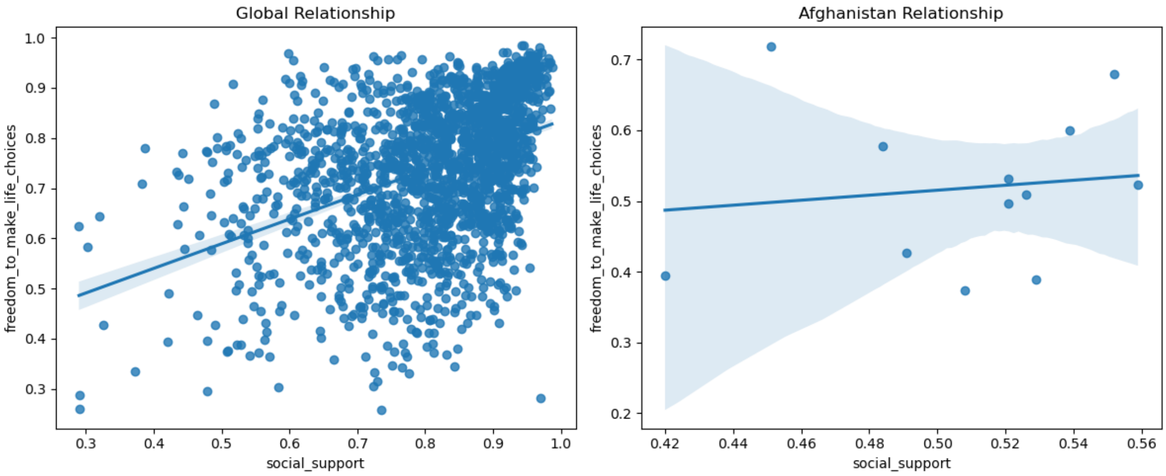
In [128…

```
sns.regplot(
    data=afg_df,
    x="log_gdp_per_capita",
    y="life_ladder"
)

plt.title("GDP vs Happiness (Afghanistan)")
plt.show()
```




In [129…

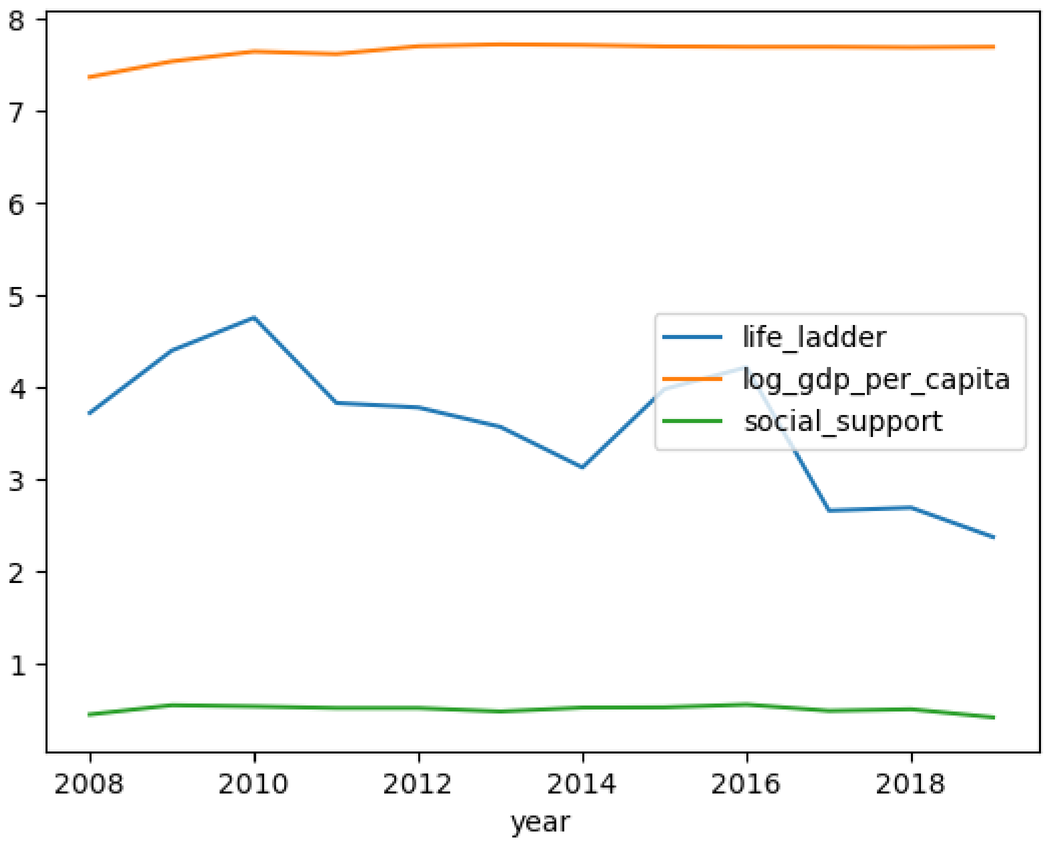
```
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
sns.regplot(
data=df,
x="social_support",
y="freedom_to_make_life_choices"
)
plt.title("Global Relationship")
plt.subplot(1,2,2)
sns.regplot(
data=afg_df,
x="social_support",
y="freedom_to_make_life_choices"
)
plt.title("Afghanistan Relationship")
plt.tight_layout()
plt.show()
```




comparing the relation graph between Globally and Afghanisthan

In [130…
```
afg_df.sort_values("year").plot(
x="year",
y=["life_ladder", "log_gdp_per_capita", "social_support"]
)
```

Out[130… <Axes: xlabel='year'>




Happiness in Afghanistan fluctuates over time, while GDP and social support show gradual trends, indicating that happiness is influenced by multiple factors beyond economic growth.

In [131…
```
df
```

Out[131...

| | country_name | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_ |
|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | |
| **...** | ... | ... | ... | ... | ... | |
| **1920** | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 | |
| **1921** | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 | |
| **1922** | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 | |
| **1923** | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 | |
| **1924** | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 | |

1925 rows × 12 columns

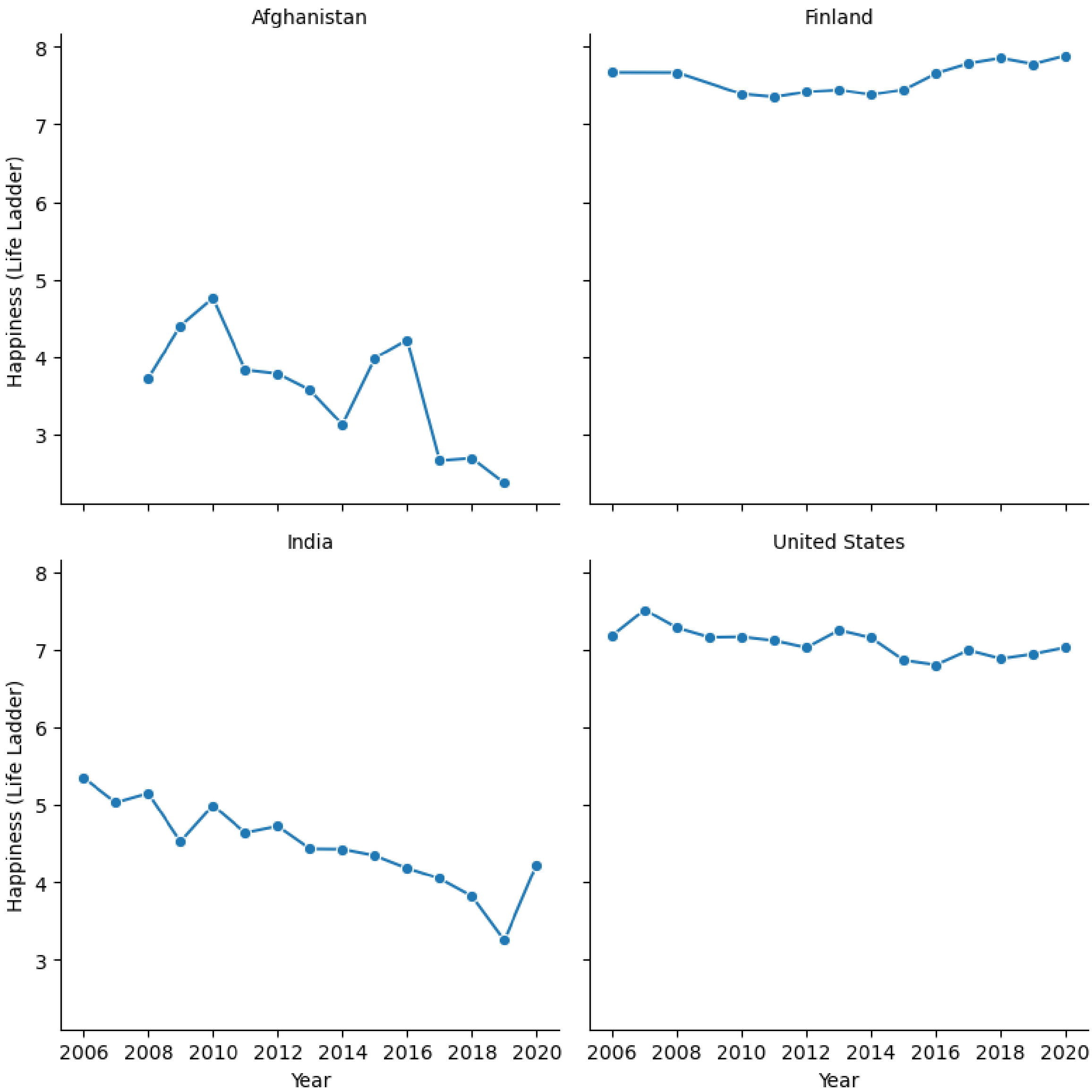
In [132...

```
afg_df["life_ladder"].std()
df["life_ladder"].std()
```

Out[132... 1.1185439019174808

In [133...

```
countries = ["Afghanistan", "India", "Finland", "United States"]

subset = df[df["country_name"].isin(countries)]

g = sns.FacetGrid(subset, col="country_name", col_wrap=2, height=4)

g.map(sns.lineplot, "year", "life_ladder", marker="o")

g.set_titles("{col_name}")
g.set_axis_labels("Year", "Happiness (Life Ladder)")

plt.tight_layout()
plt.show()
```




Finland leads in happiness with stable trends, the United States remains high with slight variation, India shows gradual improvement, and Afghanistan has the lowest and most unstable happiness levels.Developed countries tend to have higher and more stable happiness levels, while developing or conflict-affected countries show lower and more fluctuating trends.

In [134…

```
top10 = (
    df.groupby("country_name")["life_ladder"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

print(top10)
```

```
country_name
Denmark        7.680400
Finland        7.597154
Switzerland    7.548300
Norway         7.512400
Netherlands    7.466286
Iceland        7.446500
Canada         7.376333
Sweden         7.369467
New Zealand    7.310286
Australia      7.282071
Name: life_ladder, dtype: float64
```

In [135…

```python
country_mean = df.groupby("country_name")["life_ladder"].mean()

rank = country_mean.rank(ascending=False)

india_rank = rank["India"]

print("India's Rank:", int(india_rank))
```

```
India's Rank: 126
```

rank of india in happiness

In [136…

```python
country_mean
```

Out[136…

```
country_name
Afghanistan    3.594667
Albania        5.019385
Algeria        5.379286
Angola         4.420250
Argentina      6.310133
                 ...
Venezuela      6.019867
Vietnam        5.315923
Yemen          3.912250
Zambia         4.551714
Zimbabwe       3.882600
Name: life_ladder, Length: 164, dtype: float64
```

In [137…

```python
country_mean = df.groupby("country_name")["life_ladder"].mean()
last_country = country_mean.idxmin()
print("Last Country:", last_country)
```

```
Last Country: South Sudan
```

the last country in Happiness

In [138…

```python
df_2020 = df[df['year'] == 2020]

df_2020 = df_2020.sort_values(by='life_ladder', ascending=False)

df_2020['rank'] = range(1, len(df_2020) + 1)

argentina_rank = df_2020[df_2020['country_name'] == 'Argentina']

print(argentina_rank[['country_name', 'life_ladder', 'rank']])
```

```
   country_name  life_ladder  rank
50    Argentina        5.901    47
```

In [139…
```
top10 = (
df.groupby("country_name")["life_ladder"]
.mean()
.sort_values(ascending=False)
.head(10)
)

print(top10)
```

```
country_name
Denmark        7.680400
Finland        7.597154
Switzerland    7.548300
Norway         7.512400
Netherlands    7.466286
Iceland        7.446500
Canada         7.376333
Sweden         7.369467
New Zealand    7.310286
Australia      7.282071
Name: life_ladder, dtype: float64
```

In [140…
```
gdp_mean = df.groupby("country_name")["log_gdp_per_capita"].mean()
rank = gdp_mean.rank(ascending=False)
india_rank = rank["India"]
print("India's Rank:", int(india_rank))
```

```
India's Rank: 122
```

In [141…
```
india_dt= df[df["country_name"]=="India"]
```

In [142…
```
india_dt
```

Out[142...

| | country_name | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_e |
|---|---|---|---|---|---|---|
| **726** | India | 2006 | 5.348 | 8.145 | 0.707 | |
| **727** | India | 2007 | 5.027 | 8.204 | 0.569 | |
| **728** | India | 2008 | 5.146 | 8.220 | 0.684 | |
| **729** | India | 2009 | 4.522 | 8.281 | 0.653 | |
| **730** | India | 2010 | 4.989 | 8.349 | 0.605 | |
| **731** | India | 2011 | 4.635 | 8.387 | 0.553 | |
| **732** | India | 2012 | 4.720 | 8.428 | 0.511 | |
| **733** | India | 2013 | 4.428 | 8.478 | 0.553 | |
| **734** | India | 2014 | 4.424 | 8.538 | 0.621 | |
| **735** | India | 2015 | 4.342 | 8.604 | 0.610 | |
| **736** | India | 2016 | 4.179 | 8.673 | 0.614 | |
| **737** | India | 2017 | 4.046 | 8.730 | 0.607 | |
| **738** | India | 2018 | 3.818 | 8.779 | 0.638 | |
| **739** | India | 2019 | 3.249 | 8.818 | 0.561 | |
| **740** | India | 2020 | 4.225 | 8.703 | 0.617 | |

In [143...

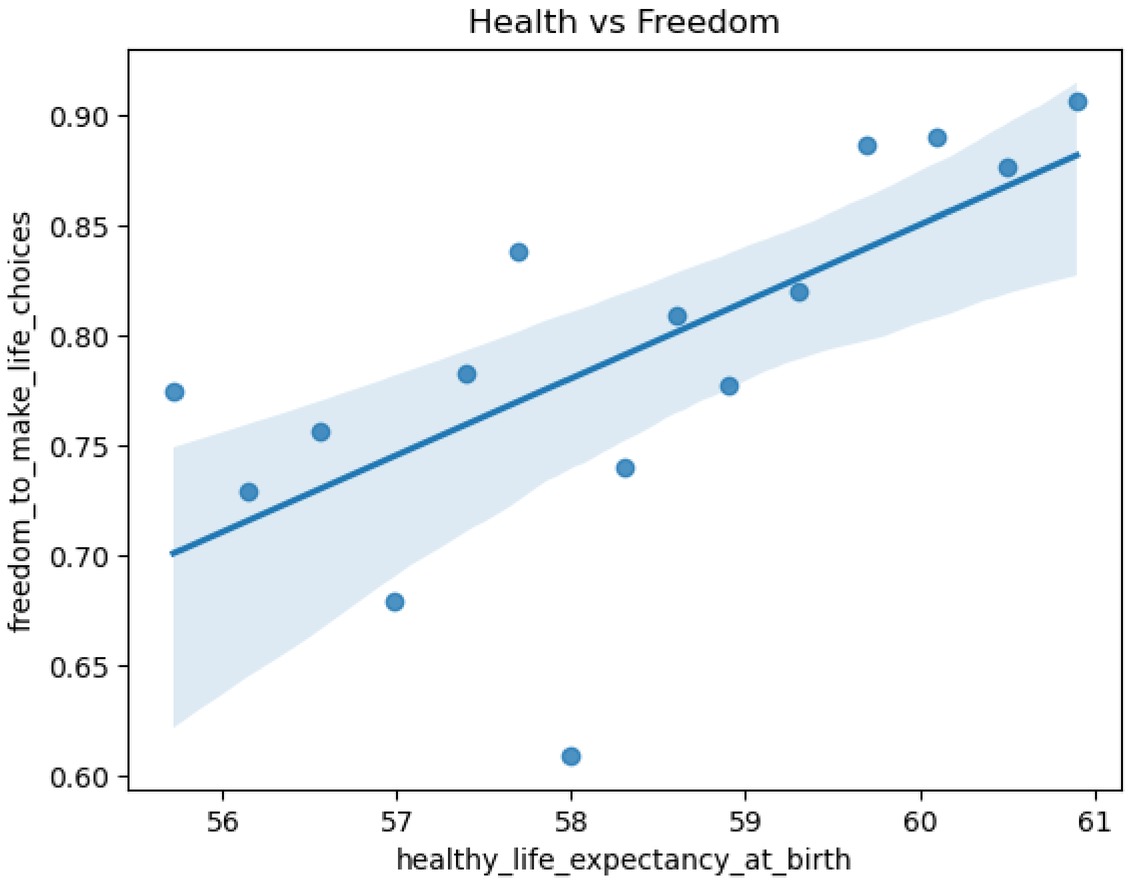
```
india_dt.mean(numeric_only=True)
```

Out[143...

```
year                               2013.000000
life_ladder                           4.473200
log_gdp_per_capita                    8.489133
social_support                        0.606867
healthy_life_expectancy_at_birth     58.320000
freedom_to_make_life_choices          0.791467
generosity                            0.017807
perceptions_of_corruption             0.828467
positive_affect                       0.686667
negative_affect                       0.307533
dtype: float64
```

In [144...

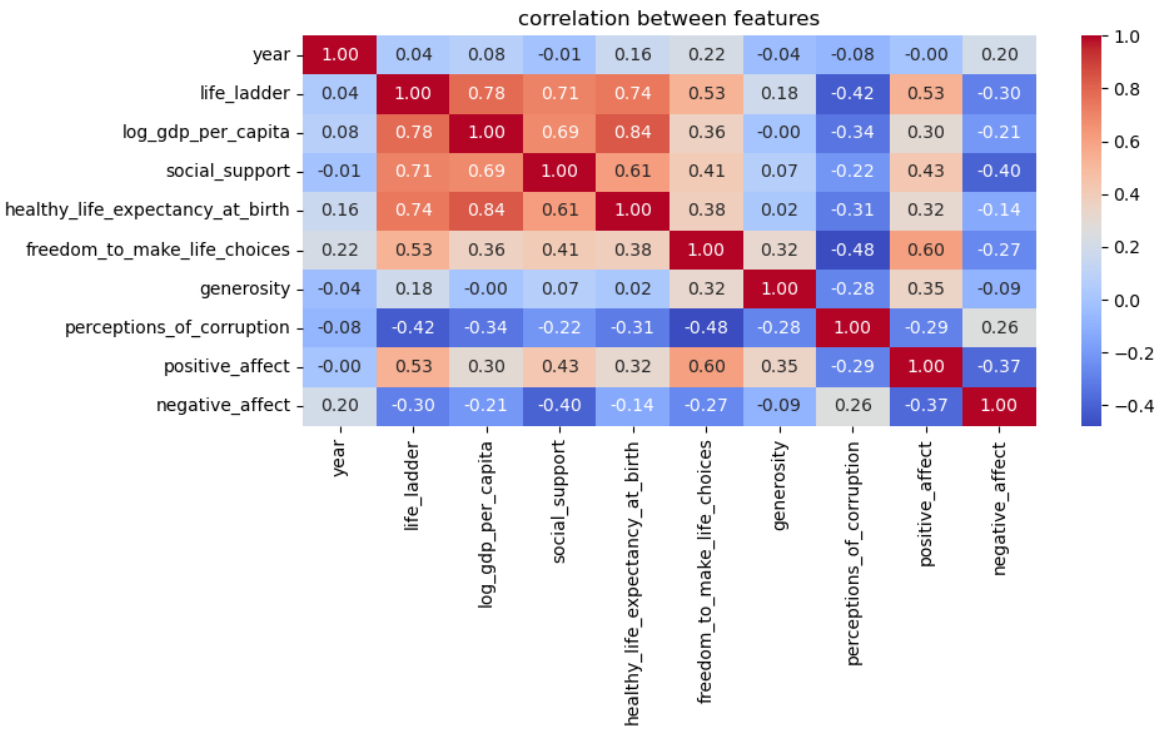
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.regplot(
    data=india_dt,
    x="healthy_life_expectancy_at_birth",
    y="freedom_to_make_life_choices"
)

plt.title("Health vs Freedom")
plt.show()
```




in health vs freedom directly proportional

In [145…

```
plt.figure(figsize=(10,6))

sns.heatmap(df.corr(numeric_only=True),annot=True,fmt=".2f",cmap="coolwarm")

plt.title("correlation between features")
plt.tight_layout()
plt.show()
```



### Insight

🔹 Strong Positive Relationships (+):log_gdp_per_capita ↔ healthy_life_expectancy_at_birth : ~0.98, year ↔ log_gdp_per_capita : ~0.98, year ↔ healthy_life_expectancy_at_birth : ~1.00, life_ladder ↔ perceptions_of_corruption : ~0.74

🔹 Strong Negative Relationships (−):life_ladder ↔ negative_affect : ~ -0.91, life_ladder ↔ log_gdp_per_capita : ~ -0.93, life_ladder ↔ year : ~ -0.89, perceptions_of_corruption ↔ log_gdp_per_capita : ~ -0.85, perceptions_of_corruption ↔ year : ~ -0.83

🔹 Moderate Relationships:life_ladder ↔ social_support : ~0.30, life_ladder ↔ freedom_to_make_life_choices : ~ -0.57, generosity ↔ negative_affect : ~0.68, freedom_to_make_life_choices ↔ log_gdp_per_capita : ~0.67

🔹 Weak / No Relationships:life_ladder ↔ positive_affect : ~0.06, positive_affect ↔ most variables : weak correlation

Happiness (life_ladder) in India is strongly negatively correlated with negative emotions and time, while economic factors like GDP and life expectancy are strongly positively correlated with each other.

In [146…
```
df = df.drop("happiness_level", axis=1)
```

In [147…
```
df
```

Out[147…

| | country_name | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_ |
|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | |
| **...** | ... | ... | ... | ... | ... | |
| **1920** | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 | |
| **1921** | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 | |
| **1922** | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 | |
| **1923** | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 | |
| **1924** | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 | |

1925 rows × 11 columns

### Data analysis over continents

In [148…
```
pip install pycountry pycountry-convert
```

```
Requirement already satisfied: pycountry in c:\users\intern\anaconda3\lib\site-pa
ckages (26.2.16)
Requirement already satisfied: pycountry-convert in c:\users\intern\anaconda3\lib
\site-packages (0.7.2)
Requirement already satisfied: pprintpp>=0.3.0 in c:\users\intern\anaconda3\lib\s
ite-packages (from pycountry-convert) (0.4.0)
Requirement already satisfied: pytest>=3.4.0 in c:\users\intern\anaconda3\lib\sit
e-packages (from pycountry-convert) (8.4.2)
Requirement already satisfied: pytest-mock>=1.6.3 in c:\users\intern\anaconda3\li
b\site-packages (from pycountry-convert) (3.15.1)
Requirement already satisfied: pytest-cov>=2.5.1 in c:\users\intern\anaconda3\lib
\site-packages (from pycountry-convert) (7.1.0)
Requirement already satisfied: repoze.lru>=0.7 in c:\users\intern\anaconda3\lib\s
ite-packages (from pycountry-convert) (0.7)
Requirement already satisfied: wheel>=0.30.0 in c:\users\intern\anaconda3\lib\sit
e-packages (from pycountry-convert) (0.45.1)
Requirement already satisfied: colorama>=0.4 in c:\users\intern\anaconda3\lib\sit
e-packages (from pytest>=3.4.0->pycountry-convert) (0.4.6)
Requirement already satisfied: iniconfig>=1 in c:\users\intern\anaconda3\lib\site
-packages (from pytest>=3.4.0->pycountry-convert) (2.1.0)
Requirement already satisfied: packaging>=20 in c:\users\intern\anaconda3\lib\sit
e-packages (from pytest>=3.4.0->pycountry-convert) (25.0)
Requirement already satisfied: pluggy<2,>=1.5 in c:\users\intern\anaconda3\lib\si
te-packages (from pytest>=3.4.0->pycountry-convert) (1.5.0)
Requirement already satisfied: pygments>=2.7.2 in c:\users\intern\anaconda3\lib\s
ite-packages (from pytest>=3.4.0->pycountry-convert) (2.19.2)
Requirement already satisfied: coverage>=7.10.6 in c:\users\intern\anaconda3\lib
\site-packages (from coverage[toml]>=7.10.6->pytest-cov>=2.5.1->pycountry-conver
t) (7.13.5)
Note: you may need to restart the kernel to use updated packages.
```

In [149…

```python
import pycountry
import pycountry_convert as pc
```

In [150…

```python
def get_continent(country_name):
    try:
        country = pycountry.countries.lookup(country_name)
        country_code = country.alpha_2
        continent_code = pc.country_alpha2_to_continent_code(country_code)

        continent_map = {
            "AF": "Africa",
            "AS": "Asia",
            "EU": "Europe",
            "NA": "North America",
            "SA": "South America",
            "OC": "Oceania"
        }

        return continent_map.get(continent_code)

    except:
        return "Unknown"

df["continent"] = df["country_name"].apply(get_continent)
```

In [151…

```python
df["continent"]
```

Out[151…
```
0          Asia
1          Asia
2          Asia
3          Asia
4          Asia
          ...
1920    Africa
1921    Africa
1922    Africa
1923    Africa
1924    Africa
Name: continent, Length: 1925, dtype: object
```

In [152…
```
df
```

Out[152…

| | country_name | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_ |
|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | |
| **...** | ... | ... | ... | ... | ... | |
| **1920** | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 | |
| **1921** | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 | |
| **1922** | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 | |
| **1923** | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 | |
| **1924** | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 | |

1925 rows × 12 columns

In [153…
```
df["continent"].value_counts()
```

Out[153…
```
continent
Asia             512
Europe           505
Africa           440
North America    170
South America    149
Unknown          121
Oceania           28
Name: count, dtype: int64
```

In [154…
```
continent_avg = df.groupby("continent")["life_ladder"].mean().sort_values(ascend

print(continent_avg)
```

```
continent
Oceania          7.296179
Europe           6.184624
North America    6.167147
South America    6.055221
Asia             5.277594
Unknown          5.213950
Africa           4.352920
Name: life_ladder, dtype: float64
```

In [155…

```
df.groupby("continent")[[
"life_ladder",
"log_gdp_per_capita",
"social_support",
"healthy_life_expectancy_at_birth"
]].mean()
```

Out[155…

| continent | life_ladder | log_gdp_per_capita | social_support | healthy_life_expectancy_at_birt |
|---|---|---|---|---|
| **Africa** | 4.352920 | 8.117064 | 0.714275 | 54.00136 |
| **Asia** | 5.277594 | 9.333158 | 0.787189 | 64.09269 |
| **Europe** | 6.184624 | 10.348536 | 0.893014 | 69.23459 |
| **North America** | 6.167147 | 9.464838 | 0.847948 | 65.67894 |
| **Oceania** | 7.296179 | 10.669286 | 0.950143 | 72.56071 |
| **South America** | 6.055221 | 9.517479 | 0.868591 | 66.33919 |
| **Unknown** | 5.213950 | 9.323828 | 0.795074 | 60.61149 |

In [156…

```
df[df["continent"] == "Unknown"]["country_name"].unique()
```

Out[156…

```
array(['Congo (Brazzaville)', 'Congo (Kinshasa)',
       'Hong Kong S.A.R. of China', 'Ivory Coast', 'Kosovo',
       'North Cyprus', 'Palestinian Territories', 'Russia',
       'Somaliland region', 'Swaziland', 'Taiwan Province of China',
       'Turkey'], dtype=object)
```

In [157…

```
df["country_name"] = df["country_name"].replace({
"United States": "United States of America",
"South Korea": "Korea, Republic of",
"North Korea": "Korea, Democratic People's Republic of",
"Russia": "Russian Federation",
"Vietnam": "Viet Nam",
"Iran": "Iran, Islamic Republic of",
"Syria": "Syrian Arab Republic",
"Tanzania": "Tanzania, United Republic of",
"Venezuela": "Venezuela, Bolivarian Republic of",
"Bolivia": "Bolivia, Plurinational State of"
})
```

In [158…

```
df["continent"] = df["country_name"].apply(get_continent)
```

In [159…
```
df["continent"].value_counts()
```

Out[159…
```
continent
Europe           520
Asia             512
Africa           440
North America    170
South America    149
Unknown          106
Oceania           28
Name: count, dtype: int64
```

In [160…
```
df = df[df["continent"] != "Unknown"]
```

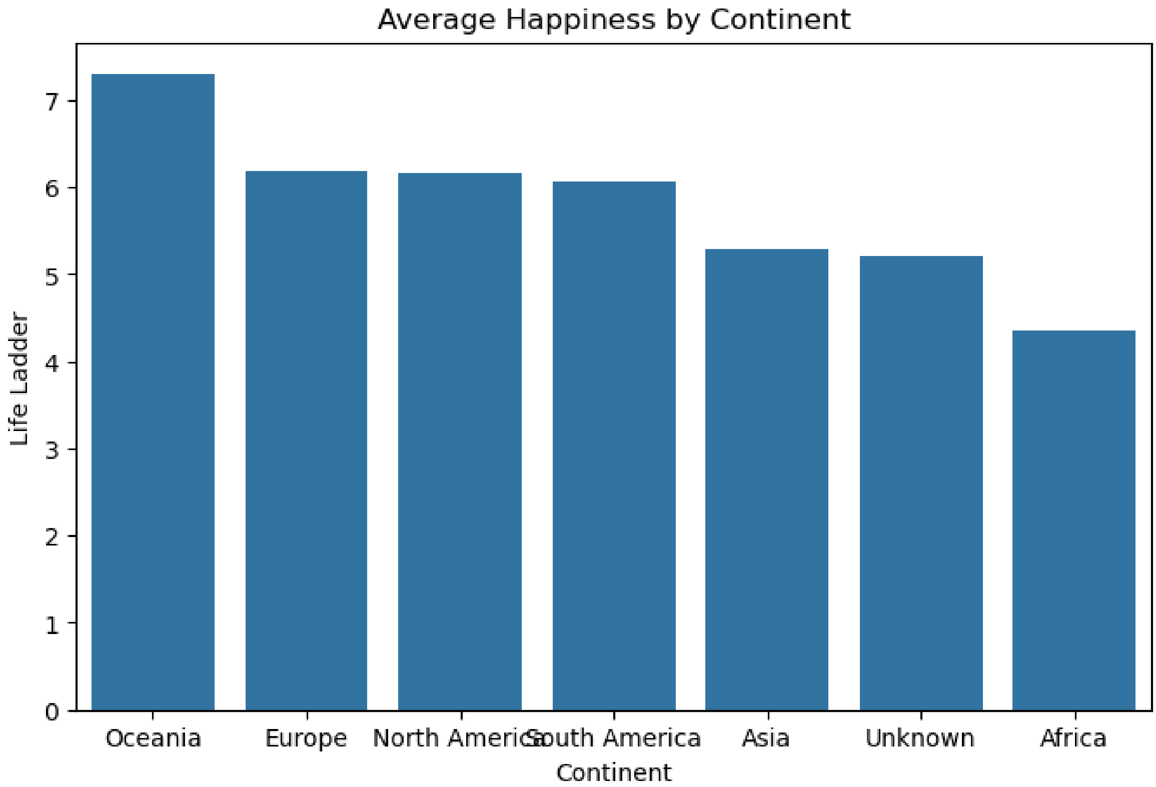
In [161…
```
df
```

Out[161…

| | country_name | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_ |
|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | |
| **...** | ... | ... | ... | ... | ... | |
| **1920** | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 | |
| **1921** | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 | |
| **1922** | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 | |
| **1923** | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 | |
| **1924** | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 | |

1819 rows × 12 columns

In [162…
```
plt.figure(figsize=(8,5))
sns.barplot(
x=continent_avg.index,
y=continent_avg.values
)
plt.title("Average Happiness by Continent")
plt.xlabel("Continent")
plt.ylabel("Life Ladder")
plt.show()
```




Oceania ranks highest in happiness, while Africa ranks lowest, showing regional inequality in well-being.

**Analysis Description**

This analysis explores global happiness trends using a dataset containing indicators such as GDP per capita, social support, health, freedom, generosity, and corruption perceptions. The data was first cleaned by handling missing values and standardizing column names for consistency.

A new feature, continent, was created to enable region-wise analysis. The study examines relationships between key variables and happiness scores using visualizations like bar charts, heatmaps, density plots, and line graphs.

The analysis compares happiness levels across continents, identifies top and bottom-performing countries, and studies trends over time. It also investigates how economicand social factors influence happiness both globally and within specific regions.

# KEY QUESTIONS:

**1.Which country is most/least happy?**

Denmark is the most happy country, while countries like Afghanistan (and in some cases South Sudan) show the lowest happiness levels.

This indicates that countries with strong economic conditions, better health, and higher social support tend to have higher happiness scores, whereas countries facing economic challenges and instability tend to have lower happiness levels.

In [163…
```
top10 = (
df.groupby("country_name")["life_ladder"]
.mean()
.sort_values(ascending=False)
.head(10)
)
print(top10)
```

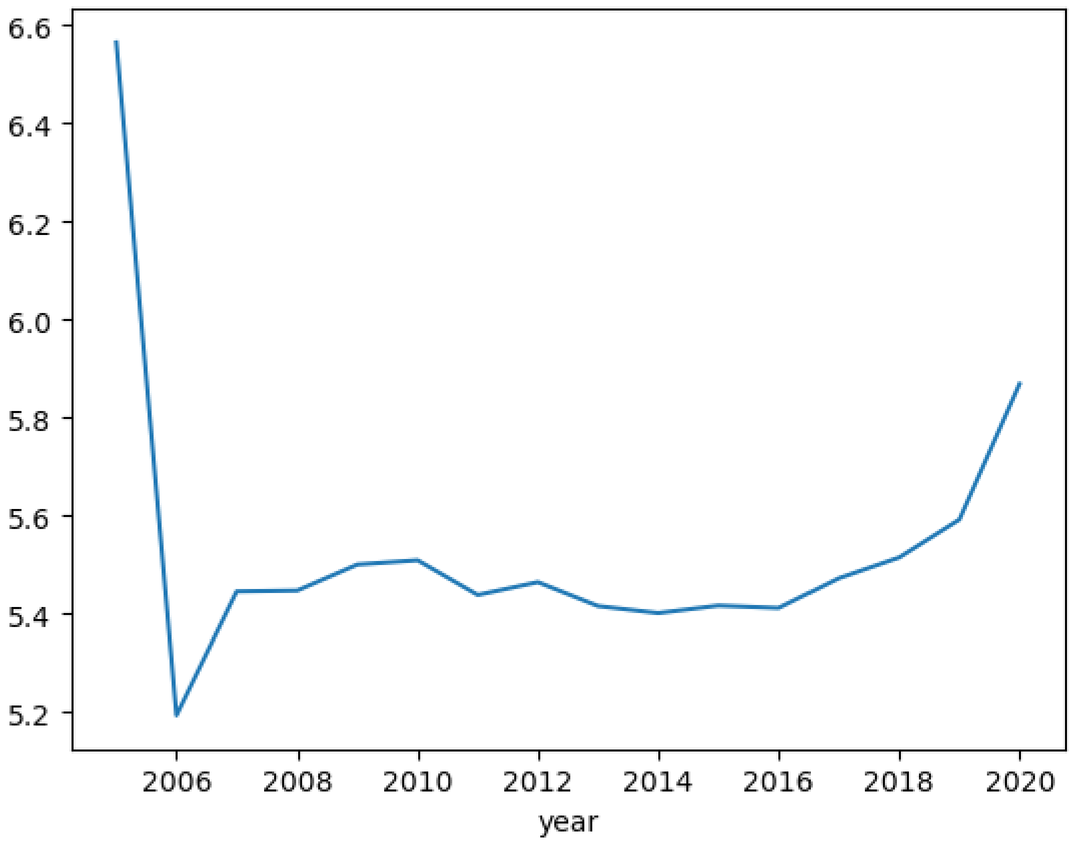
```
country_name
Denmark        7.680400
Finland        7.597154
Switzerland    7.548300
Norway         7.512400
Netherlands    7.466286
Iceland        7.446500
Canada         7.376333
Sweden         7.369467
New Zealand    7.310286
Australia      7.282071
Name: life_ladder, dtype: float64
```

#### 2.How does happiness change over time?

The results show that happiness levels remain relatively stable over time, with only small fluctuations. There is no extreme increase or decrease globally, but slight variations may occur due to economic conditions, global events, or regional differences.

In [164…
```
df.groupby("year")["life_ladder"].mean().plot()
```

Out[164… <Axes: xlabel='year'>




#### 3.Does GDP affect happiness?

The analysis shows a strong positive relationship, meaning that countries with higher GDP per capita generally have higher happiness levels.

This suggests that better economic conditions improve living standards, access to resources, and overall well-being, which contributes to higher happiness.

In [165…

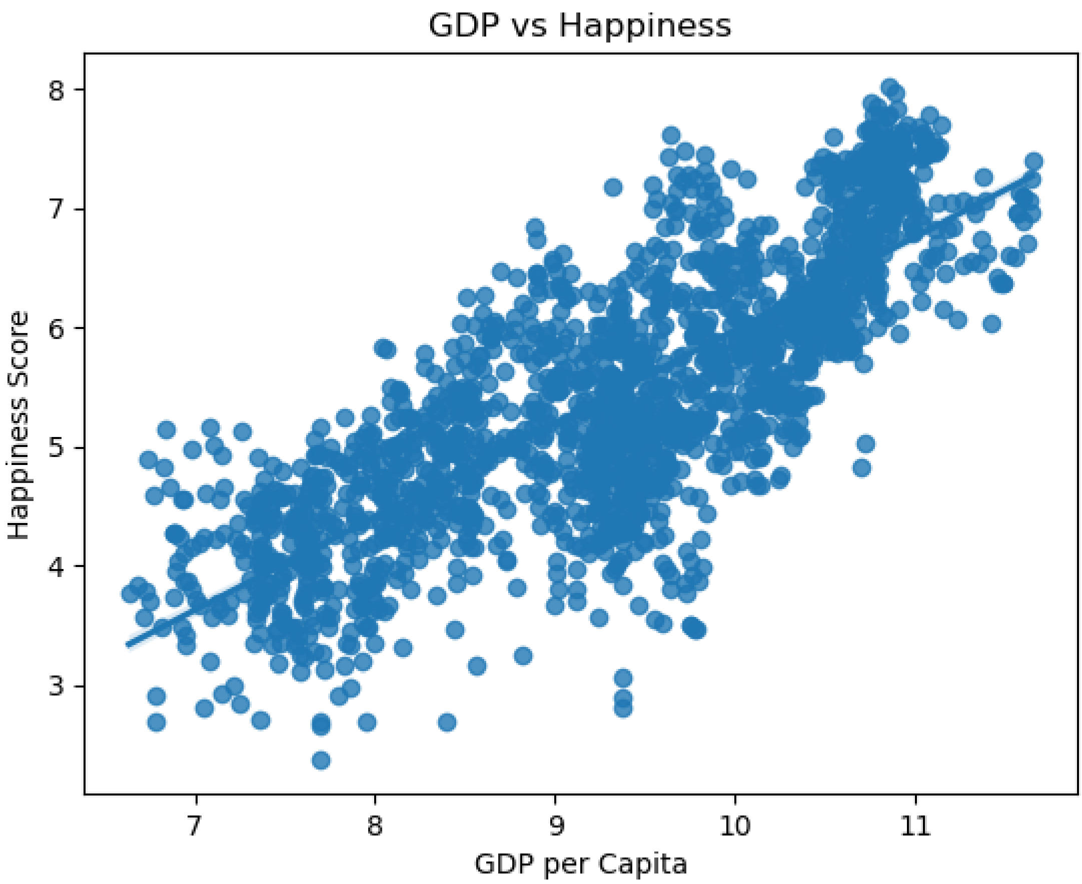
```
df["log_gdp_per_capita"].corr(df["life_ladder"])
```

Out[165…

```
np.float64(0.7910312633651448)
```

In [166…

```
sns.regplot(x='log_gdp_per_capita', y='life_ladder', data=df)
plt.xlabel("GDP per Capita")
plt.ylabel("Happiness Score")
plt.title("GDP vs Happiness")
plt.show()
```



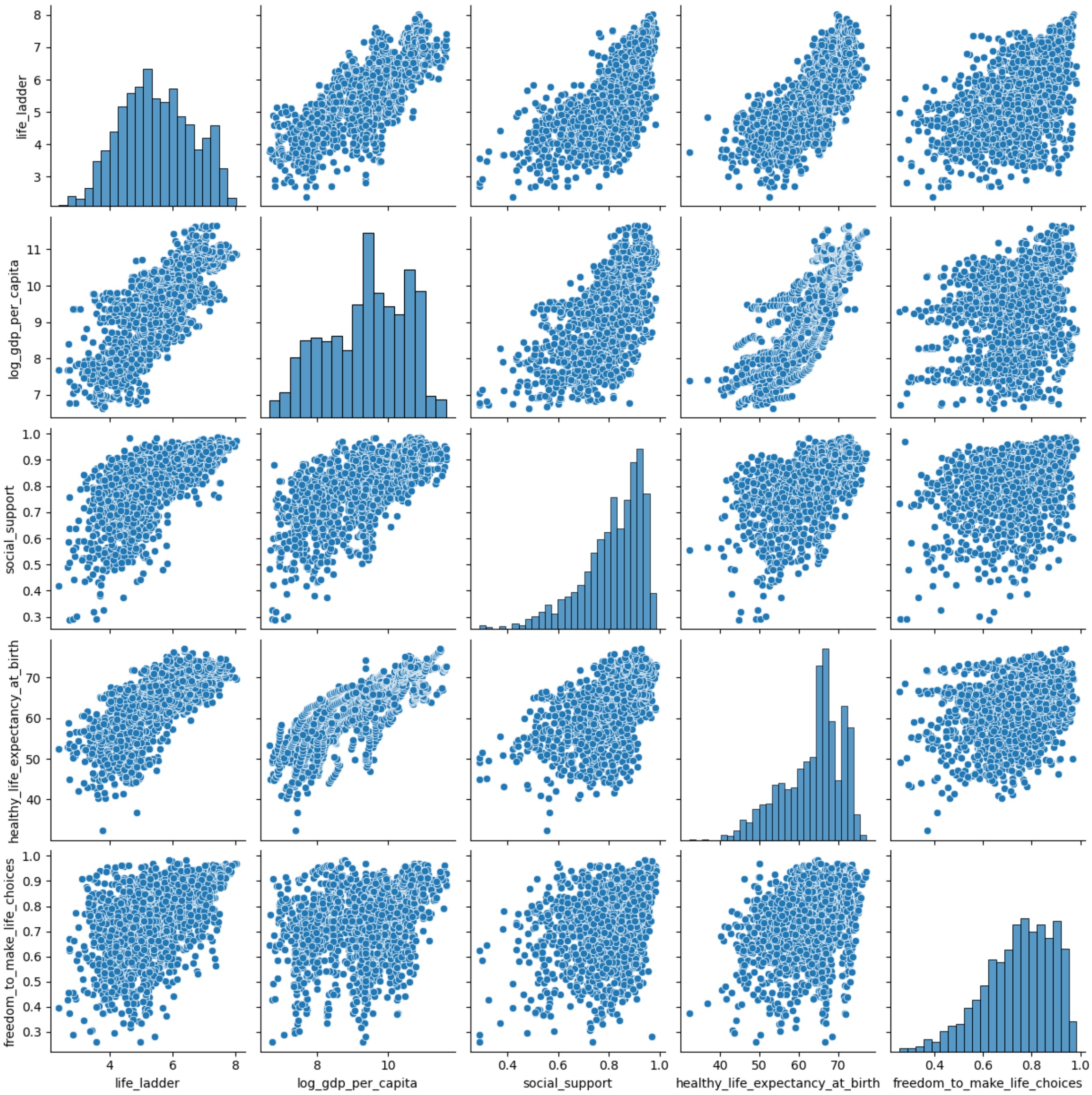

**4.Can we predict happiness using other features?**

Yes, happiness can be predicted using other features such as GDP per capita, social support, health, freedom, generosity, and corruption perception.

In [167…

```
sns.pairplot(
df[[
"life_ladder",
"log_gdp_per_capita",
"social_support",
"healthy_life_expectancy_at_birth",
"freedom_to_make_life_choices"
]]
```

```
)
plt.show()
```



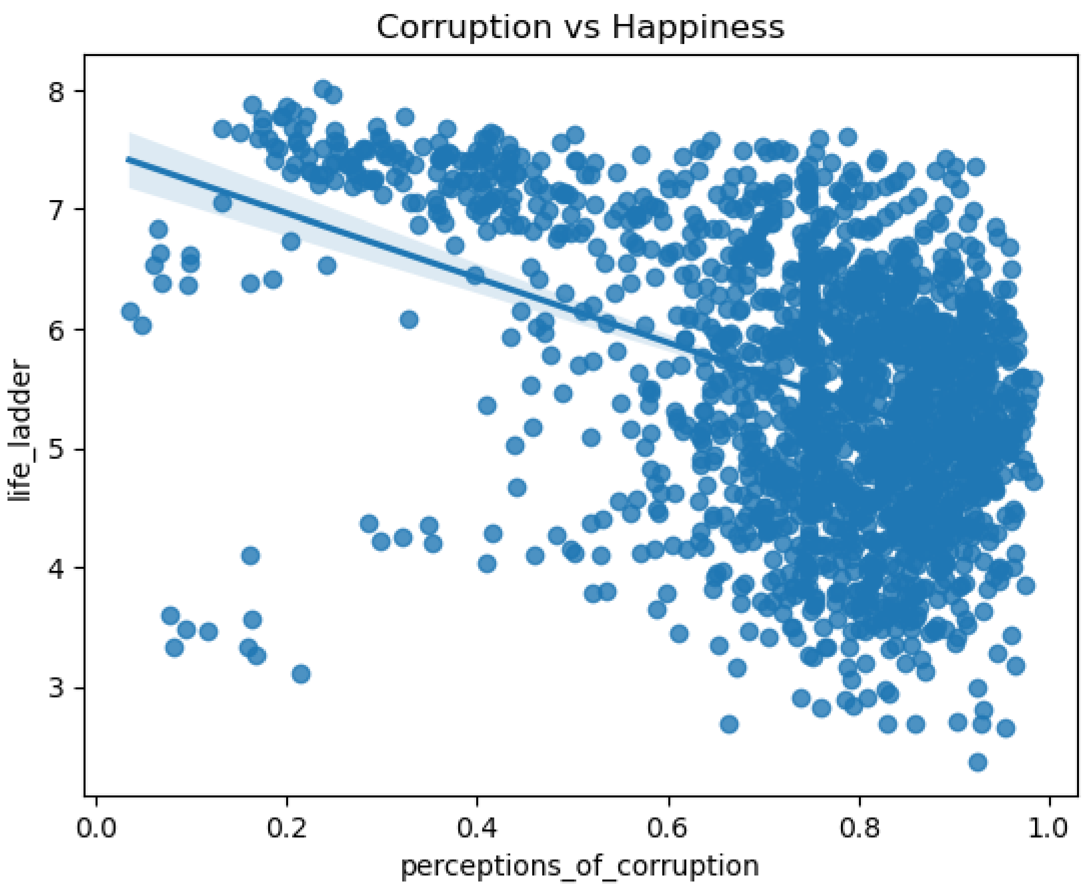

### 5.How does corruption affect happiness?

The analysis shows a negative relationship between corruption and happiness. This means that as the perception of corruption increases, the happiness level tends to decrease.

In [168...
```
df["perceptions_of_corruption"].corr(df["life_ladder"])
```

Out[168...
```
np.float64(-0.4364015579924175)
```

In [169...
```
sns.regplot(data=df, x="perceptions_of_corruption", y="life_ladder")
plt.title("Corruption vs Happiness")
plt.show()
```



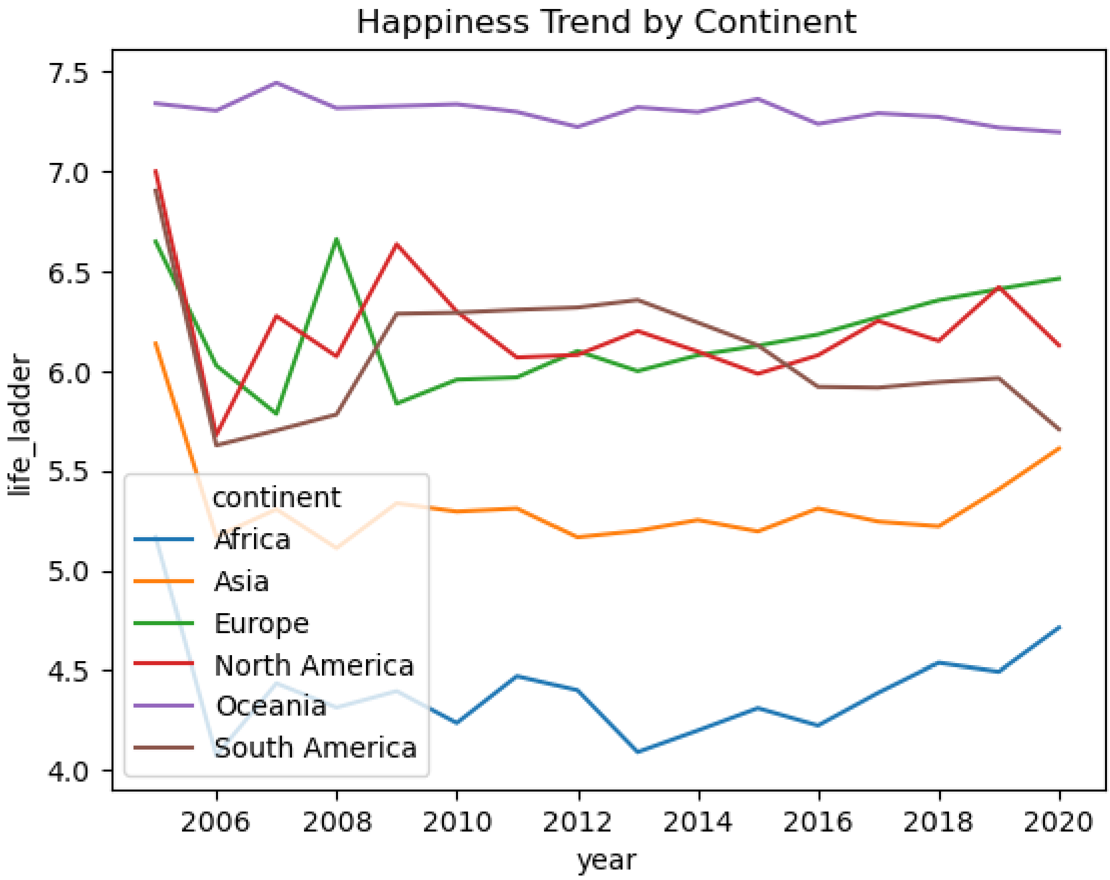

**6.How does happiness vary across continents?**

Happiness levels vary significantly across different continents. By grouping the data by continent and calculating the average life_ladder score, clear regional differences can be observed.

👉 Oceania shows the highest happiness levels, indicating strong economic conditions, health, and social support. 👉 Europe and North America also have high happiness scores, reflecting developed economies and better quality of life. 👉 South America has moderate happiness levels, slightly lower but relatively stable. 👉 Asia shows mixed results due to variation between developed and developing countries. 👉 Africa has the lowest happiness levels, likely due to lower GDP, health challenges, and limited resources.

In [171…

```
continent_trend = df.groupby(["year", "continent"])["life_ladder"].mean().reset_
sns.lineplot(
data=continent_trend,
x="year",
y="life_ladder",
hue="continent"
)
plt.title("Happiness Trend by Continent")
plt.show()
```


Happiness Trend by Continent
life_ladder
year
continent
Africa
Asia
Europe
North America
Oceania
South America
7.5
7.0
6.5
6.0
5.5
5.0
4.5
4.0
2006
2008
2010
2012
2014
2016
2018
2020

In [ ]: